CompTIA

DATA+ DA0-001

COURSE SLIDES · WSQ

WSQ - CompTIA Certified Data+ Training

WSQ Course Code: TGS-2024049212

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code generated from TPG.

3

Scan with Singpass

Scan the QR code from your Singpass App and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Data analytics, data engineering, statistics, business intelligence and applied machine learning.

DELIVERS
WSQ courses on data analytics, CompTIA certifications, Python, AI and cloud technologies.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and current role.
- How you work with data today — spreadsheets, SQL, BI tools, or none of these yet.
- The one data question in your job you wish you could answer confidently.
- Whether you intend to sit the CompTIA Data+ (DA0-001) certification exam.

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HOUSEKEEPING

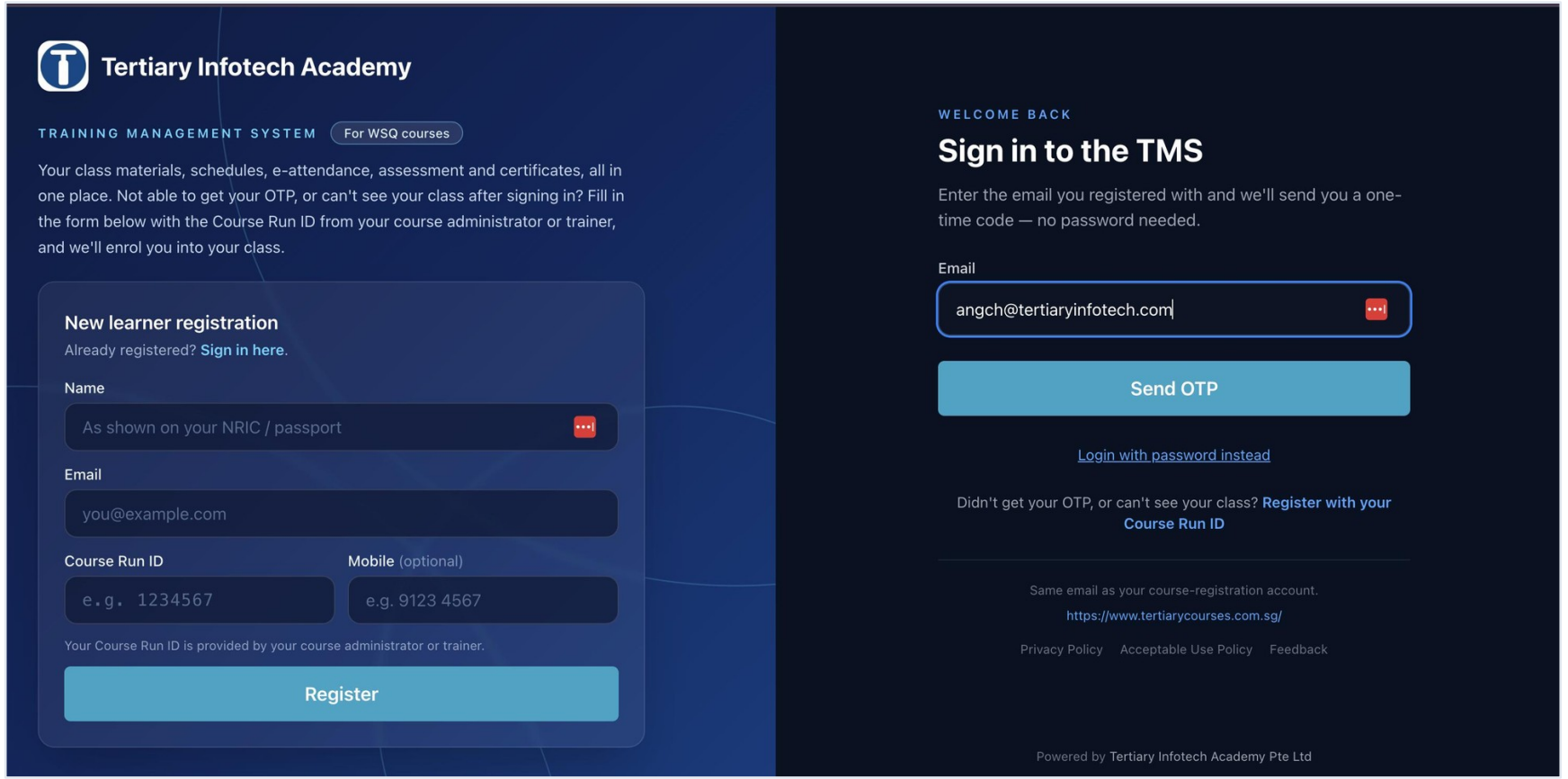
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively in class — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual and return from breaks on time.
6. Step out quietly for calls or breaks.
7. 75% attendance is required for funding.
8. Complete both assessments to be certified Competent.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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LMS / TMS · lms-tms.tertiaryinfotech.com

Download Your Course Material

1

1 · Go to the LMS portal

Open https://lms-tms.tertiaryinfotech.com in your browser.

2

2 · Log in

Sign in with the account e-mail you used to register for this course.

3

3 · Open this course

Select 'WSQ - CompTIA Certified Data+ Training' (TGS-2024049212).

4

4 · Download

Download the Trainer Slides, Learner Guide and Lesson Plan (PDF).

5

5 · Get the labs

Clone or download the lab repository from GitHub — link on the next slide.

6

6 · Keep them open

You may use these materials during the open-book assessment.

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Skills Framework Alignment

1

TSC Title

Data Analytics

2

TSC Code

ATP-PIN-3001-1.1

3

K1 · K2

Tools and techniques for data collection, mining, cleaning and analysis; statistical principles.

4

K3 · K4

Principles of communicating data effectively; procedures to extract and process data sets.

5

A1 · A2 · A3

Integrate multiple datasets; conduct data mining to uncover trends; derive actionable insights.

6

A4 · A5

Recognise sequential and sub-graph patterns; present analytics outputs visually.

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Integrate datasets

Integrate multiple datasets to extract and process data efficiently. (K4, A1)

2

LO2 · Mine for trends

Conduct data mining to uncover and analyze trends using analysis techniques. (K1, A2)

3

LO3 · Analyse statistically

Perform statistical data analysis to derive actionable insights. (K2, A3)

4

LO4 · Communicate visually

Present analytical outputs visually to communicate data effectively for decision-making. (K3, A5)

5

LO5 · Link the variables

Recognize and interpret sequential patterns to establish linkages between variables. (A4)

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FIVE DOMAINS · 15 HANDS-ON LABS

Course Outline — Five Exam Domains

1

Domain 1 — Data Concepts and Environments (20%)

Database types · data structures · file extensions · data types · data sources · infrastructure · tools · AI concepts

2

Domain 2 — Data Acquisition and Preparation (22%)

Data integration · queries · exploration · missing values · duplication · outliers · cleansing · parsing · formatting

3

Domain 3 — Data Analysis (24%)

Descriptive statistics · inferential methods · analysis types · communicating results · troubleshooting

4

Domain 4 — Visualization and Reporting (20%)

Chart selection · maps · tables · design elements · dashboards · report types · validation

5

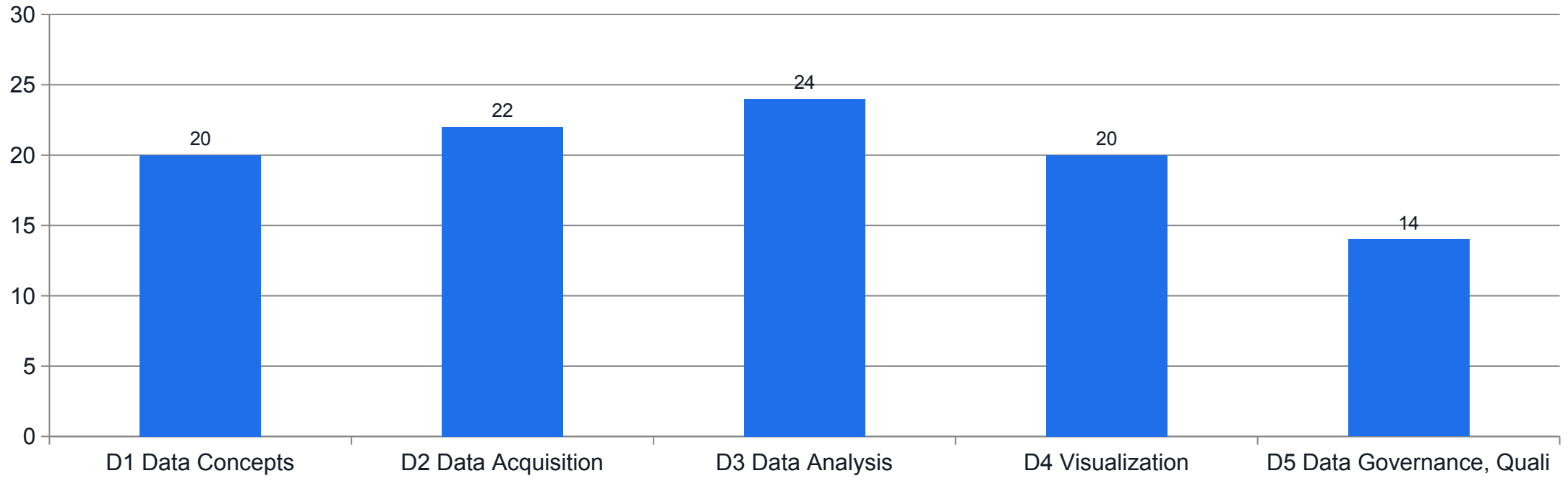
Domain 5 — Data Governance, Quality and Controls (14%)

Documentation · versioning · lineage · compliance · retention · privacy · encryption · masking · quality assurance

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EXAM BLUEPRINT · DA0-001

How the CompTIA Data+ Exam Is Weighted



WHAT THE DATA SHOWS

Data Analysis (24%) and Data Acquisition & Preparation (22%) together make up 46% of the exam — which is why Days 2 and 3 carry the heaviest lab load.

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SCHEDULE · 1 OF 2

Lesson Plan — Days 1 to 3

Days 1 and 2

- Day 1 — Data Concepts, Schemas and Environments
  - Digital Attendance (AM) · Introductions
  - Learning Outcomes · Course Outline
  - Domain 1: Data Concepts (Labs 1–3)
  - Lunch · Digital Attendance (PM)
- Day 2 — Data Acquisition, Exploration and Preparation
  - Domain 2: Acquisition & Preparation
  - Labs 4–7 · profiling, parsing, joins

Day 3

- Day 3 — Data Analysis — Descriptive and Inferential Statistics
  - Domain 3: Data Analysis
  - Descriptive statistics (Lab 8)
  - Hypothesis testing (Lab 9)
  - Correlation and regression (Lab 10)
- Every day
  - 9:00 am – 6:00 pm · 8 training hours
  - 1-hour lunch; tea breaks within

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SCHEDULE · 2 OF 2

Lesson Plan — Days 4 and 5

Day 4

- Day 4 — Visualization, Dashboards and Reporting
  - Domain 4: Visualization and Reporting
  - Chart selection (Lab 11)
  - KPI dashboard and validation (Lab 12)
  - Report types and design principles
  - Lunch · Digital Attendance (PM)

Day 5

- Day 5 — Data Governance, Quality, Controls and Final Assessment
  - Domain 5: Governance, Quality, Controls
  - Labs 13–15 · protect, assess risk, automate QA
  - Revision across all five domains
  - Course Feedback and TRAQOM Survey
  - Digital Attendance (Assessment)
  - Final Assessment 4:00–6:00 PM
  - WA (1 hr) then PP (1 hr) · open book

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TOOLS YOU'LL USE · ALL BROWSER-BASED

Your Lab Environment

1

Killercoda Ubuntu Playground

A free, browser-based Ubuntu terminal with Python 3, pandas and SQLite — no local install needed. · https://killercoda.com/playgrounds/scenario/ubuntu

2

RegexLab

Real-time regular-expression tester used to build and validate the parsing patterns that clean messy text fields. · https://alfredang.github.io/regexgenerator/

3

PCAP Analyzer

Browser-based packet-capture analyser — the course's worked example of machine/log data as a data source. · https://alfredang.github.io/pcapanalyzer/

4

IP Calculator

Subnet calculator used to derive structured numeric fields from raw network address data. · https://alfredang.github.io/ipcalculator/

5

Cybersecurity Threat Simulator

Risk-scoring and data-leakage simulator used in the governance domain to reason about classification and protection. · https://alfredang.github.io/cybersecuritysimulator/

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GITHUB · ALL 15 LABS

Access the Hands-On Labs

Repository: https://github.com/tertiarycourses/TGS-2024049212-CompTIA-Certified-Data-Training

OPTION A · CLONE

- Open the Killercoda Ubuntu playground.
- git clone https://github.com/tertiarycourses/TGS-2024049212-CompTIA-Certified-Data-Training.git
- cd into the labs folder and start Lab 1.

OPTION B · DOWNLOAD ZIP

- Open the repository page on GitHub.
- Click the green Code button → Download ZIP.
- Unzip it and open the labs folder.

NO INSTALL NEEDED

Every lab runs free in the browser on the Killercoda Ubuntu playground — https://killercoda.com/playgrounds/scenario/ubuntu

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BEFORE YOU START

Briefing for Assessment

1

Do · Clear your desk

Place phones and other materials under the table or on the floor.

2

Don't · No recording

No photos or recording of assessment scripts.

3

Don't · No discussion

Work individually — no discussion during the assessment.

4

Do · Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

Don't · No correction fluid

No liquid paper or correction tape may be used.

6

Do · Stop on time

Assessment scripts are collected when the time is up.

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FINAL ASSESSMENT

Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) · 1 hour · individual · summative · open book.

2

Practical Performance (PP)

Scenario-based hands-on data tasks · 1 hour · individual · summative · open book.

3

Open book means

Course slides, the Learner Guide and other approved materials only.

4

Eligibility

A minimum of 75% attendance is required to be eligible for assessment and funding.

5

Result

You are assessed Competent (C) or Not Yet Competent (NYC) on each instrument.

6

Appeals

An appeal process is available if you wish to contest an assessment outcome.

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ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM attendance

Scan the QR on the LMS

2

Assessment attendance

Scan the SSG QR

3

Sit WA then PP

Open book · 1 hour each

4

Submit on the LMS

Upload your answers

5

Sign the record

Assessment Summary Record

REMEMBER

All five steps are mandatory for WSQ funding — missing the TRAQOM digital attendance or the course feedback survey can invalidate your claim.

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WSQ FUNDING

Criteria for Funding

1

Attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessment

Complete both assessment components and be assessed as 'Competent'.

3

Digital attendance

Scan the SSG QR code for AM, PM and Assessment on every training day.

4

Course feedback

Complete the mandatory course feedback / TRAQOM survey on the LMS.

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01

DOMAIN 01 · 20% OF THE EXAM

Data Concepts and Environments

Database types · data structures · file extensions · data types · data sources · infrastructure · tools · AI concepts

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WHAT THIS DOMAIN COVERS

Domain 1 — Key Concepts

1. **Schemas are designed first**
   A data schema describes both how data is organised and how tables relate to one another — it is designed before a single row is written.

2. **Relational vs non-relational**
   Relational databases store data in linked tables and are queried with SQL; non-relational (NoSQL) stores trade fixed schemas for scale and flexibility.

3. **Normalise, then denormalise**
   Normalisation (1NF–5NF) removes redundancy; denormalisation deliberately re-introduces it to make analytical queries fast.

4. **OLTP vs OLAP**
   OLTP systems are built for many small real-time transactions; OLAP systems are built for long, complex analytical queries.

5. **Warehouse, mart, lake, lakehouse**
   Data warehouses, data marts, data lakes and lakehouses each hold data at a different level of structure, cost and query flexibility.

6. **Structure and file formats**
   Data is structured, semi-structured or unstructured, and is carried in file formats such as CSV, TSV, JSON, XML, XLSX and Parquet.

7. **Cloud, containers and AI**
   Modern data platforms run on-premise, in the cloud or in containers, and AI/ML, NLP and RPA are now standard parts of the data toolchain.

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DATA SCHEMAS

A schema is a plan, not an afterthought.

A data schema describes both the organisation of the data and the relationships between tables. Database engineers design the schema BEFORE they create the system — because changing it later means migrating every row.

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DATABASE TYPES

Relational vs Non-Relational Databases

Characteristic	Relational (SQL)	Non-Relational (NoSQL)
Structure	Tables of rows and columns	Documents, key-value, column or graph
Schema	Fixed, defined before loading	Flexible, schema-on-read
Query language	SQL — standard across vendors	Vendor APIs, GraphQL, MongoDB query
Field limits	Typed, length-constrained fields	Store as much as you want per key
Scaling	Vertical — a bigger server	Horizontal — add more nodes
Best for	Transactions, integrity, joins	Huge volume, varied or unstructured data
Examples	MySQL, MariaDB, PostgreSQL, Aurora	MongoDB, Redis, Cassandra, Neo4j

WHEN IT MATTERS Remember for the exam: SQL = relational. NoSQL or GraphQL = non-relational.

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NoSQL FAMILIES

The Four Non-Relational Database Types

1

Document-oriented

Stores each record as a self-describing XML or JSON document. Example: MongoDB, CouchDB.

2

Key-value store

Stores each value against a unique key — the simplest and fastest model. Example: Redis, DynamoDB.

3

Column-oriented

Stores data down columns instead of across rows, which makes analytical scans very fast. Example: Cassandra, HBase.

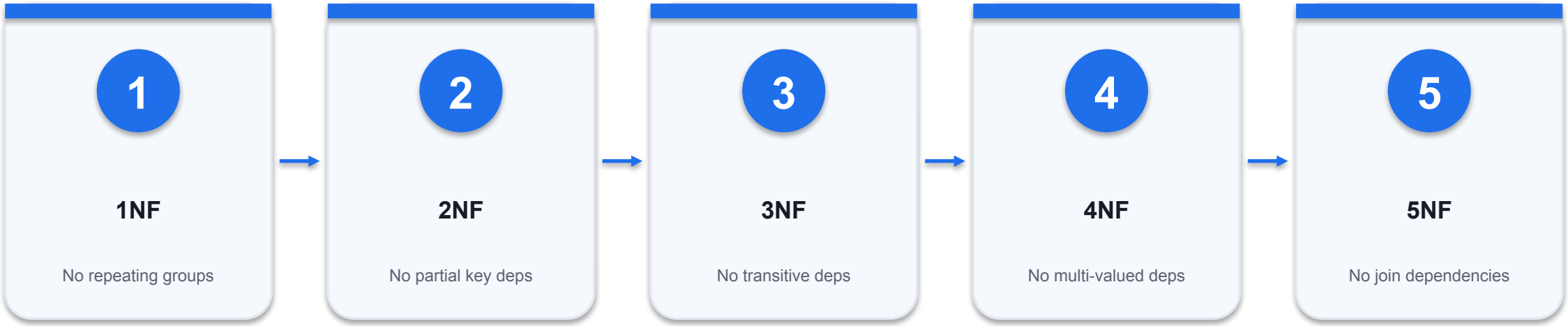
4

Graph store

Stores entities as nodes and relationships as edges, so linked queries are cheap. Example: Neo4j, Amazon Neptune.

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Normalisation — 1NF to 5NF



THE GOAL

Normalisation removes redundancy so an update happens in exactly one place. Each form builds on the one before it.

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DESIGN DECISION

Should I Normalise or Denormalise?



NORMALISE → OLTP

Many small writes, integrity matters most. Remove every redundancy so an update touches one row. This is your operational database.

DENORMALISE → OLAP

Few huge reads, speed matters most. Deliberately repeat data to avoid expensive joins. This is your warehouse, mart or reporting layer.

Data warehousing, mining, analysis and visualization all work on deliberately denormalised data.

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Keys and Relationships

1

Primary key

A unique identifier for a record that cannot contain duplicates or nulls. Every table needs one.

2

Foreign key

A primary key from one table referenced by another — this is what creates the relationship.

3

One-to-one

One record relates to exactly one record in the other table (person ↔ passport).

4

One-to-many

One record relates to many records in another table (customer → many orders).

5

Many-to-many

Many records relate to many others, resolved with a junction table (students ↔ courses).

6

Referential integrity

Guarantees every foreign key value actually exists in the parent table — no orphans.

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REFERENTIAL INTEGRITY

Cascade your changes, or orphan your data.

When a parent record is updated or deleted, ON UPDATE CASCADE and ON DELETE CASCADE push that change through every dependent record. Without them, deleting a customer leaves their orders pointing at nothing — the classic integrity failure you will build and break in Lab 1.

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OLTP vs OLAP — Two Different Jobs

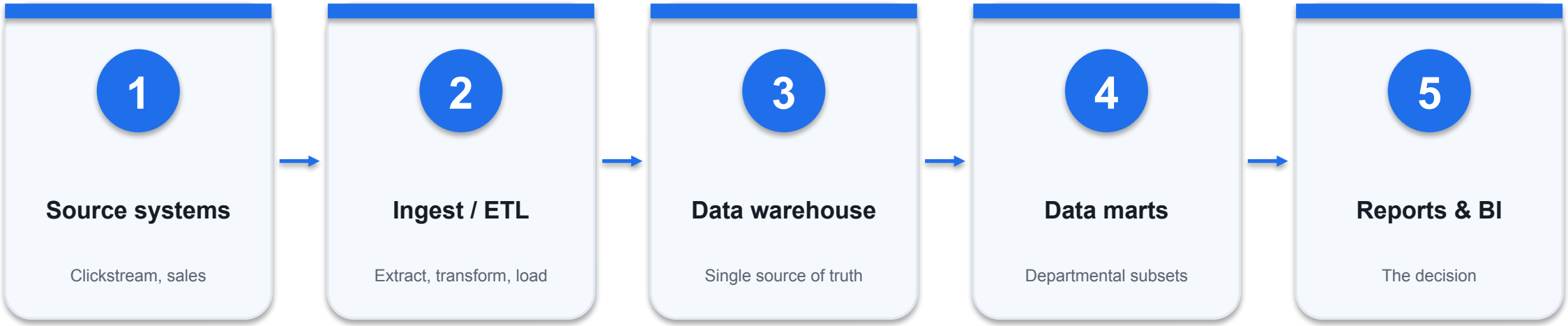
Characteristic	OLTP	OLAP
Purpose	Run the business	Analyse the business
Transaction type	Insert, update, delete, small query	Long, complex analytical query
Volume per query	One or a few rows	Millions of rows
Users	Many concurrent operational users	Fewer analysts and reports
Data model	Normalised (3NF)	Denormalised (star / snowflake)
Example	Point-of-sale, booking system	Data warehouse, BI dashboard

WHEN IT MATTERS Choosing the wrong system is a classic cause of performance problems — a heavy analytical query on an OLTP database will lock up the business.

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DATA WAREHOUSE ARCHITECTURE

From Source System to Insight



WHY IT MATTERS

The warehouse combines many source systems into one consistent, efficient place — the single source of truth every report can agree on.

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Star vs Snowflake Schema

Aspect	Star Schema	Snowflake Schema
Shape	Fact table with one ring of dimensions	Dimensions branch into further dimensions
Dimension depth	One level — denormalised	Multiple levels — normalised
Joins per query	Fewer	More
Query speed	Faster	Slower
Storage	More redundancy, more space	Less redundancy, less space
Maintenance	Simpler to query, more to update	Harder to query, easier to update

WHEN IT MATTERS A fact table holds the measures and the keys; a dimension table holds the descriptive attributes.

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STORAGE ARCHITECTURES

Warehouse · Mart · Lake · Lakehouse

1

Data warehouse

Structured, schema-on-write, single source of truth. Built for reliable, repeatable reporting.

2

Data mart

A subset of the warehouse for one department, so their queries do not slow anyone else down.

3

Data lake

A centralised repository holding structured AND unstructured data cheaply, schema-on-read.

4

Data lakehouse

Queries data in place on the lake using an added schema layer — lake economics, warehouse queries.

5

OLAP cube

A pre-aggregated multi-dimensional structure that answers expected queries almost instantly.

6

Choosing

Structured and well-understood → warehouse. Varied, raw and high-volume → lake or lakehouse.

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Slowly Changing Dimensions

1

The problem

A dimension table holds descriptive metadata — but names, addresses and categories change over time.

2

Type 1

Overwrite the old value. Simple, but the history is gone forever — you can never query the old name.

3

Type 2

Add a new row with effective dates. Keeps the COMPLETE history, at the cost of more rows.

4

Type 3

Add a 'previous value' column. Keeps current and prior only — a compromise between Type 1 and Type 2.

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Quantitative vs Qualitative Data

	Quantitative (numeric)	Qualitative (categorical)
Definition	Measured with numbers	Grouped by a quality or category
Sub-type 1	Discrete — countable whole values (orders)	Nominal — named, no order (city, colour)
Sub-type 2	Continuous — any value in a range (weight)	Ordinal — ordered categories (small/medium/large)
Maths allowed	Full arithmetic — sum, mean, SD	Counts and modes only
Typical chart	Histogram, line, scatter	Bar chart, pie chart

WHEN IT MATTERS Getting this wrong is the root cause of nonsense analysis — you cannot take the mean of a postal code.

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Data Field Types

1

Character / text / string

Alphanumeric. A number stored as text CANNOT be used in arithmetic until it is converted.

2

Numeric — integer

Whole numbers only. Use for counts, IDs and quantities.

3

Numeric — decimal / float

Fractional values. Beware floating-point rounding in financial calculations.

4

Currency

A special numeric type fixed at two decimal places, representing money.

5

Date / datetime

A calendar date, optionally with a time. Decide the format and the century convention up front.

6

Boolean

Exactly two values — true/false, yes/no, 1/0. The basis of all logical operations.

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CONVERTING DATA

Data is dynamic, not static.

Just because a value was stored as text does not mean it must stay text. Converting a text field to an integer, decimal or date is a routine, expected part of preparation — and it is what makes calculation possible.

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DATA STRUCTURES

Structured · Semi-Structured · Unstructured

	Structured	Semi-structured	Unstructured
Organisation	Fixed rows and columns	Self-describing tags/keys	None
Schema	Defined before loading	Carried inside the file	No schema at all
Examples	SQL tables, CSV	JSON, XML, HTML, e-mail	Text, images, audio, video
Storage	Relational database	Document store, lake	Blob (Azure) / Bucket (AWS)
Query effort	Low — query directly	Moderate — parse then query	High — extract before you can query

WHEN IT MATTERS You measure this difference yourself in Lab 2 — the same record in all three forms.

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FILE EXTENSIONS

File Formats and Extensions

1

.CSV

Comma-separated values — the universal flat file. One record per line.

2

.TSV / .TAB

Tab-separated. Safer than CSV when the data itself contains commas.

3

.TXT

Plain text with any delimiter you choose — pipe, semicolon, fixed width.

4

.JSON

JavaScript Object Notation. Self-describing, nestable, the de facto format for APIs.

5

.XML

Extensible Markup Language. Tag-based, built to TRANSFER data rather than display it.

6

.XLSX / .PARQUET

Spreadsheet workbook; and Parquet, a compressed columnar format built for analytics.

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Data Languages and Markup

1

SQL

The structured query language. SELECT chooses fields, FROM chooses tables, WHERE filters rows.

2

HTML

Hypertext Markup Language — tags that tell a browser how to DISPLAY a page. Semi-structured.

3

XML

Similar tags to HTML, but the purpose is to TRANSFER data between systems, not display it.

4

JSON

Arrays and key-value objects — lighter than XML and native to JavaScript and web APIs.

5

Python

General-purpose language dominant in data preparation, analysis and machine learning.

6

R

A language built specifically for statistical computing and graphics.

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Infrastructure — Where Data Lives

1

On-premise

Servers you own and run in your own facility. Maximum control, maximum capital cost.

2

Cloud

Infrastructure rented on demand from AWS, Azure or Google Cloud. Elastic and operational-cost based.

3

Hybrid

Sensitive data stays on-premise; burst analytical workloads run in the cloud.

4

Storage tiers

Hot storage for active data, cold and archive tiers for cheap long-term retention.

5

Containerisation

Docker packages an application with its dependencies so it runs identically anywhere.

6

Orchestration

Kubernetes schedules and scales those containers across a cluster automatically.

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AI Concepts in the Data Toolchain

1

Machine learning

Models learn patterns from historical data to classify or predict, rather than being explicitly programmed.

2

Supervised vs unsupervised

Supervised learns from labelled examples; unsupervised finds structure such as clusters in unlabelled data.

3

Natural language processing

Lets systems parse, classify and generate human language — sentiment analysis, entity extraction, summarisation.

4

Large language models

NLP models trained at scale that generate text; increasingly used to draft queries and explain results.

5

Robotic process automation

Software robots that perform repetitive, rule-based data tasks — extraction, entry, reconciliation.

6

Where analysts meet AI

Most often in data preparation, anomaly detection, forecasting and automated report narratives.

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DATA SOURCES

Identifying Data Sources

1

Databases

Relational and NoSQL operational systems — usually the most reliable and best-documented source.

2

APIs

A defined request/response contract between systems. Pull model polls; push model notifies on change.

3

Website data

Web scraping with Selenium, BeautifulSoup or Scrapy. Fragile, and you must have permission.

4

Files

CSV, Excel, JSON and XML extracts, often the practical reality of inter-company data exchange.

5

Logs & machine data

Server, application and device telemetry — high volume, and the basis of predictive maintenance.

6

Repositories

Public and licensed datasets: data.gov, Kaggle, Pew Research, GeoPostcodes, Singapore's data.gov.sg.

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Lab 1 — Build a Relational Schema and Prove Referential Integrity

LAB 1

Explain data concepts: database types, data structures and data types (Domain 1); LO1 / K4 / A1.

You design and build a small three-table sales schema in SQLite, declare the primary and foreign keys, and then deliberately attack it with an orphan record and a cascading delete so you can see referential integrity doing its job.

YOU'LL BUILD

A normalised 3NF SQLite database (customers, products, orders) with enforced foreign keys, plus evidence of a rejected orphan insert.

TOOLCHAIN

Killercoda Ubuntu, SQLite 3, SQL DDL/DML

DONE WHEN

The three tables import 40 customers, 10 products and 120 orders. Attack 1 fails with 'FOREIGN KEY constraint failed'. Attack 2 deletes customer 1 AND cascades to their 2 orders, so the order count drops from 120 to 118. The final join returns 118 priced order lines.

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DOMAIN 01 · LAB 1

How Lab 1 Runs

1

Download this lab's dataset

step 1 of 9

2

Create the customers table

step 3 of 9

3

Create the orders fact table carrying TWO fo

step 5 of 9

4

ATTACK 1

step 7 of 9

5

Run a join across all three tables to confir

step 9 of 9

YOU'LL PRODUCE

A normalised 3NF SQLite database (customers, products, orders) with enforced foreign keys, plus evidence of a rejected orphan insert.

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Lab 1 — Build a Relational Schema and Prove Referential Integrity — Procedure 1 of 3

1

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

```
$ mkdir -p ~/dataplus/lab1 && cd ~/dataplus/lab1;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-01-build-a-relational-schema-and-prove-referential-inte;
   for f in customers.csv products.csv orders.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

Turn foreign-key enforcement ON — SQLite leaves it off by default, which is the single most common cause of orphaned rows.

```
$ PRAGMA foreign_keys = ON;
```

3

Create the customers table — the columns must match customers.csv exactly, or .import silently drops data.

```
$ CREATE TABLE customers (customer_id INTEGER PRIMARY KEY, first_name TEXT NOT NULL, last_name TEXT NOT NULL,
   email TEXT UNIQUE, region TEXT, joined DATE);
```


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Lab 1 — Build a Relational Schema and Prove Referential Integrity — Procedure 2 of 3

4 Create the products table — note REAL for currency and the CHECK constraint enforcing domain integrity.

```
$ CREATE TABLE products (product_id INTEGER PRIMARY KEY, name TEXT NOT NULL, unit_price REAL CHECK (unit_price
   >= 0));
```

5 Create the orders fact table carrying TWO foreign keys with ON DELETE CASCADE.

```
$ CREATE TABLE orders (order_id INTEGER PRIMARY KEY, customer_id INTEGER REFERENCES customers(customer_id) ON
   DELETE CASCADE, product_id INTEGER REFERENCES products(product_id), qty INTEGER NOT NULL, order_date DATE);
```

6 Import the three CSV extracts into the tables you just created.

```
$ sqlite3 sales.db ".mode csv" ".import --skip 1 customers.csv customers" ".import --skip 1 products.csv
   products" ".import --skip 1 orders.csv orders" "SELECT COUNT(*) FROM customers; SELECT COUNT(*) FROM products;
   SELECT COUNT(*) FROM orders;"
```


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Lab 1 — Build a Relational Schema and Prove Referential Integrity — Procedure 3 of 3

7

ATTACK 1 — insert an order for a customer who does not exist, using a FREE order_id so the error you see is the foreign key firing and not a duplicate primary key. This MUST be rejected.

```
$ INSERT INTO orders VALUES (9999,999,10,1,'2025-03-03');
```

8

ATTACK 2 — delete customer 1 and watch the cascade remove their orders, leaving nothing orphaned.

```
$ DELETE FROM customers WHERE customer_id = 1; SELECT * FROM orders;
```

9

Run a join across all three tables to confirm the schema answers a real business question.

```
$ SELECT c.first_name, p.name, o.qty, o.qty*p.unit_price AS line_total FROM orders o JOIN customers c ON
   c.customer_id=o.customer_id JOIN products p ON p.product_id=o.product_id;
```


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Lab 1 — Verify Your Work

✅ Expected result

The three tables import 40 customers, 10 products and 120 orders. Attack 1 fails with 'FOREIGN KEY constraint failed'. Attack 2 deletes customer 1 AND cascades to their 2 orders, so the order count drops from 120 to 118. The final join returns 118 priced order lines.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

The orphan insert SUCCEEDED

You forgot PRAGMA foreign_keys = ON. It resets on every new connection — re-run it after reopening sqlite3.

It says UNIQUE constraint failed, not FOREIGN KEY

That order_id already exists in orders.csv. Pick an unused id (9999) so the foreign key is what fails.

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Lab 2 — Compare Structured, Semi-Structured and Unstructured Data

LAB 2

Explain data concepts: data structures, file extensions and data types; identify data sources (Domain 1); LO1 / K4.

You create the same customer record three ways — as a CSV row, as a JSON document and as free text — then measure how much work each format takes to query. This is the exam's structured vs semi-structured vs unstructured distinction, made concrete.

YOU'LL BUILD

Three files (customers.csv, customers.json, notes.txt) plus a Python script that queries each and reports the effort required.

TOOLCHAIN

Killercoda Ubuntu, Python 3, csv and json modules

DONE WHEN

The CSV and the JSON both total 2293.17 across 12 customers. The JSON also returns ['vip','repeat'] for the first record — a field the flat CSV cannot represent at all. The regex over notes.txt returns extra noise, proving unstructured text needs parsing before it can be analysed.

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How Lab 2 Runs

1. **Download this lab's dataset** — step 1 of 5
2. **Query the CSV** — step 2 of 5
3. **Query the JSON** — step 3 of 5
4. **Try to query the unstructured text** — step 4 of 5
5. **Compare the file sizes and record what each** — step 5 of 5

YOU'LL PRODUCE

Three files (customers.csv, customers.json, notes.txt) plus a Python script that queries each and reports the effort required.

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Lab 2 — Compare Structured, Semi-Structured and Unstructured Data — Procedure 1 of 2

1

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

```
$ mkdir -p ~/dataplus/lab2 && cd ~/dataplus/lab2;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-02-compare-structured-semi-structured-and-unstructured;
   for f in customers.csv customers.json notes.txt; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

Query the CSV — three lines of code, because the structure is guaranteed.

```
$ python3 -c "import csv;print(sum(float(r['spend']) for r in csv.DictReader(open('customers.csv'))))"
```

3

Query the JSON — still easy, and it carries the extra 'tags' field the CSV could not hold.

```
$ python3 -c "import json;
   d=json.load(open('customers.json'));
   print(sum(r['spend'] for r in d));
   print(d[0].get('tags'))"
```

4

Try to query the unstructured text — you need a regular expression, and it is fragile.

```
$ python3 -c "import re;
    t=open('notes.txt').read();
    print(re.findall(r'[$]?([0-9]+(?:[.][0-9]{2})?)\s*(?:dollars)?',t))"
```


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Lab 2 — Compare Structured, Semi-Structured and Unstructured Data — Procedure 2 of 2

Compare the file sizes and record what each format cost you in query effort.

5

```
$ ls -l customers.csv customers.json notes.txt
```


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Lab 2 — Verify Your Work

✅ Expected result

The CSV and the JSON both total 2293.17 across 12 customers. The JSON also returns ['vip','repeat'] for the first record — a field the flat CSV cannot represent at all. The regex over notes.txt returns extra noise, proving unstructured text needs parsing before it can be analysed.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

JSONDecodeError

customers.json did not download cleanly. Check it with head -c 80 customers.json — if it starts with '404' the URL is wrong.

The regex returns '240.50' and '89' plus junk

That is the expected lesson — unstructured text has no guarantees. Tighten the pattern in RegexLab.

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Lab 3 — Profile Machine/Log Data as a Data Source (PCAP Analyzer)

LAB 3

Identify data sources: logs, machine data and repositories; recognise infrastructure concepts (Domain 1); LO1 / A1.

Machine-generated data is one of the exam's named data sources, and it never arrives analysis-ready. You load a packet capture into the browser-based PCAP Analyzer, read the statistics it derives, and translate what you see into the data-analyst vocabulary of records, fields, dimensions and measures.

YOU'LL BUILD

A completed data-source profile of a machine-data feed: record count, field inventory, dimensions vs measures, and three analytical questions it can answer.

TOOLCHAIN

PCAP Analyzer (https://alfredang.github.io/pcapanalyzer/), any .pcap/.pcapng sample

DONE WHEN

You can state the record count and average packet size, classify at least five fields as dimension or measure, and explain why Protocol Distribution is a frequency table rather than a raw record list.

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DOMAIN 01 · LAB 3

How Lab 3 Runs

1

Open the PCAP Analyzer in your browser

step 1 of 8

2

Drag the

step 3 of 8

3

Open the Protocol Distribution panel

step 5 of 8

4

Click any single packet to inspect its field

step 7 of 8

5

Write down three business questions this fee

step 8 of 8

YOU'LL PRODUCE

A completed data-source profile of a machine-data feed: record count, field inventory, dimensions vs measures, and three analytical questions it can answer.

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Lab 3 — Profile Machine/Log Data as a Data Source (PCAP Analyzer) — Procedure 1 of 2

1. Open the PCAP Analyzer in your browser. Everything is parsed locally — nothing is uploaded.

2. **Generate a small capture on Killercoda if you do not have one, then download it to your machine.**

```
$ sudo tcpdump -i any -c 200 -w ~/sample.pcap 2>/dev/null || echo 'use the sample capture supplied by your
   trainer'
```

3. Drag the .pcap file onto the drop zone and wait for the four summary statistics to appear.

4. Record the four derived MEASURES: packet count, total bytes, capture duration and average packet size.

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Lab 3 — Profile Machine/Log Data as a Data Source (PCAP Analyzer) — Procedure 2 of 2

5. Open the Protocol Distribution panel — this is a categorical frequency table, exactly like a GROUP BY.

6. Open Top Talkers and Top Conversations — these are aggregations over a source/destination DIMENSION.

7. Click any single packet to inspect its fields, and list which are dimensions (IP, protocol) and which are measures (length).

8. Write down three business questions this feed could answer, and one it cannot — noting what extra data you would need.

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Lab 3 — Verify Your Work

✅ Expected result

You can state the record count and average packet size, classify at least five fields as dimension or measure, and explain why Protocol Distribution is a frequency table rather than a raw record list.

IF IT DOESN'T WORK

The file will not load

The analyser accepts .pcap and .pcapng only. Confirm the extension, and that the file is not zero bytes (ls -l).

tcpdump: permission denied

Prefix with sudo on Killercoda. If it is still blocked, use the sample capture your trainer provides.

The statistics look empty

A capture with zero packets produces zero rows. Re-capture with a larger -c value while browsing in another tab.

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DOMAIN 01 RECAP

Recap — Data Concepts and Environments

1. **Schemas**
   Schemas define both organisation and relationships, and are designed before the system is built.
2. **Relational vs NoSQL**
   Relational databases use SQL and fixed tables; non-relational stores trade schema for scale and flexibility.
3. **Normalisation**
   Normalisation removes redundancy for OLTP; denormalisation restores it for fast OLAP analysis.
4. **Keys & integrity**
   Primary and foreign keys plus referential integrity are what stop your data becoming orphaned.
5. **Storage architectures**
   Warehouses, marts, lakes and lakehouses trade structure, cost and query flexibility against each other.
6. **Structure & formats**
   Data is structured, semi-structured or unstructured, and arrives in CSV, JSON, XML, XLSX or Parquet.
7. **Infrastructure & AI**
   Infrastructure may be on-premise, cloud, hybrid or containerised — and AI/NLP/RPA are now part of the toolchain.

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Lunch Break

1 hour

02

DOMAIN 02 · 22% OF THE EXAM

Data Acquisition and Preparation

Data integration · queries · exploration · missing values · duplication · outliers · cleansing · parsing · formatting

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WHAT THIS DOMAIN COVERS

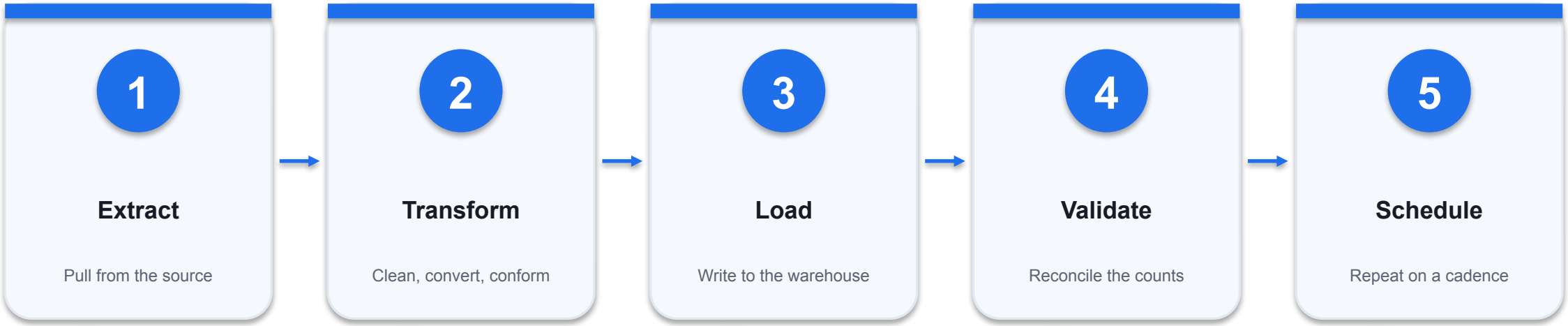
Domain 2 — Key Concepts

1. **ETL vs ELT**
   ETL transforms data before it lands in the warehouse; ELT loads it raw into a lake and transforms it later, on demand.
2. **Where data comes from**
   Data can be acquired from databases, APIs, web scraping, machine and log data, public repositories, surveys and sampling.
3. **Profile before you trust**
   Data profiling is the disciplined first pass: identify the source, the field names and types, the keys, and what the data actually contains.
4. **The four defect classes**
   The four recurring data-quality defects are missing values (NULL), duplicated or redundant records, invalid values, and outliers.
5. **Cleansing techniques**
   Cleansing techniques include filtering or imputing NULLs, deduplication, trimming invisible characters, and correcting data types.
6. **Manipulation techniques**
   Data manipulation covers recoding, derived variables, imputation, aggregation, transposing, appending, merging and parsing.
7. **Joins, filters and indexes**
   SQL joins (inner, left, right, full), filters, subqueries, indexes and execution plans are how analysts combine and optimise datasets.

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DATA ACQUISITION

The ETL Pipeline



THE RULE

In ETL the data is transformed BEFORE it lands — so what arrives in the warehouse is already conformed and trustworthy.

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INTEGRATION METHODS

ETL vs ELT — When to Use Which

Aspect	ETL	ELT
Order	Extract → Transform → Load	Extract → Load → Transform
Transform runs	In a staging/processing layer	Inside the target platform
Target	Data warehouse	Data lake / lakehouse / cloud warehouse
Schema	Schema-on-write, decided up front	Schema-on-read, decided at query time
Best when	Requirements are stable and known	You want to keep raw data for future, unknown uses
Trade-off	Clean on arrival, less flexible	Flexible, but transformation debt accumulates

WHEN IT MATTERS Use ELT when you are landing into a data lake — hold the raw data and transform it when the question is known.

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Loading Strategies and Frequency

1

Full load

Every row is loaded every time. Simple and self-correcting, but slow and expensive at scale.

2

Delta / incremental load

Only new or changed records since the last run. Fast, but you must track the watermark reliably.

3

Batch processing

Large volumes processed on a schedule — nightly, hourly. The default for reporting.

4

Real-time / streaming

Records processed as they arrive. Needed when the decision cannot wait for the batch.

5

Transaction processing

Large volumes of individual transactions handled synchronously as they occur.

6

Distributed processing

Very large datasets split across many servers so they can be processed in parallel.

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API Pull vs Push

Aspect	Pull model	Push model
Who initiates	Your system requests the data	The source notifies you when data changes
Timing	On your schedule — polling	On the source's event — webhook
Freshness	As fresh as your polling interval	Near real-time
Cost	Wasted calls when nothing changed	Efficient, but needs a listener endpoint
Failure mode	You miss changes between polls	A missed notification is lost unless retried

WHEN IT MATTERS Understanding the FREQUENCY of change in the source is what tells you which model you need.

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WHERE THE DATA COMES FROM

Acquisition Methods and Their Risks

1

Database query

Most reliable — typed, documented, joinable. Use it whenever the source offers it.

2

API

A defined contract, usually JSON over HTTPS. Watch rate limits, pagination and authentication.

3

Web scraping

Selenium, BeautifulSoup, Scrapy. FRAGILE — a layout change breaks it — and you must have permission.

4

Public repositories

data.gov.sg, data.gov, Kaggle, Pew Research, GeoPostcodes. Check the licence and the currency of the data.

5

Survey data

Bias is the enemy. Cover the whole range of responses; use a Likert scale for shades of opinion.

6

Sampling & observation

Random, systematic or stratified sampling reduces volume — but a biased sample invalidates everything downstream.

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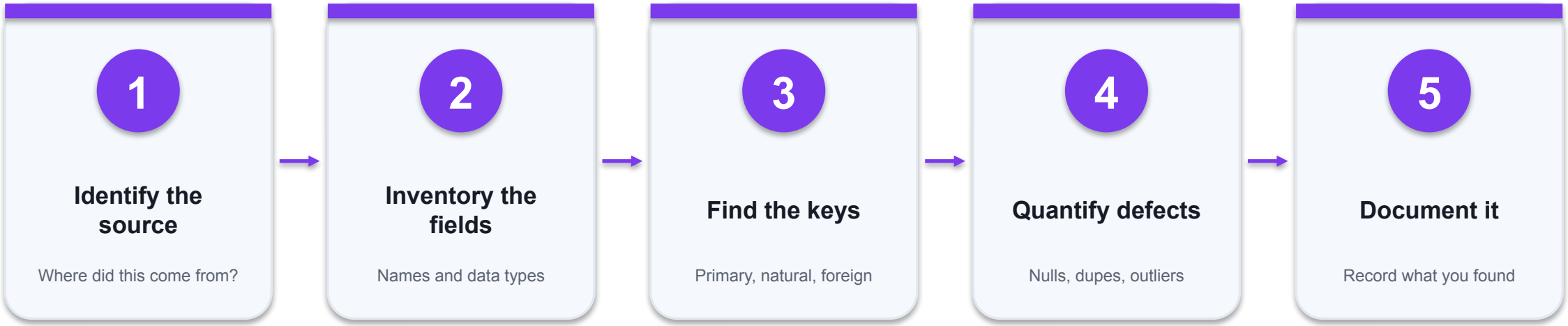
DATA PROFILING

Profile before you trust. Always.

Data profiling is the disciplined first pass over any new dataset: identify the source, the field names and types, the keys, the row count and what the values actually contain. Skipping it is how a wrong number reaches the board.

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The Data Profiling Procedure



THE OUTPUT

A written profile that says how bad the data is BEFORE you clean it — your baseline for measuring the cleaning.

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The Four Defect Classes You Must Find

1

Missing values (NULL)

Blank, NULL or N/A. Causes: not applicable, never collected, mismatched fields, incomplete survey.

2

Duplicated data

The same record repeated within one dataset. Inflates every count, sum and average.

3

Redundant data

Identical data stored in multiple places. Not always wrong — but it must be reconciled.

4

Invalid data

Wrong type, hard-coded values, invisible/non-printable ASCII characters, leading and trailing spaces.

5

Outliers

Values far outside the normal distance from the rest. May be an error — or the most important signal.

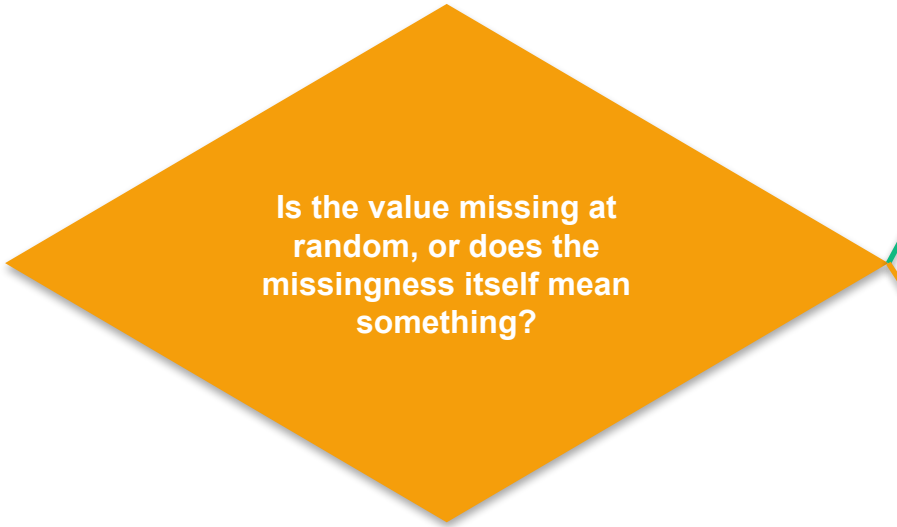
6

Specification failures

Data that does not meet the type or quality the schema requires — usually a wrong data type.

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A Value Is Missing. What Do I Do?



MISSING AT RANDOM

Impute it — substitute the mean, median or a modelled estimate — or filter the row out. Document the choice and the count.

MEANINGFUL MISSING

Keep it as NULL and treat it as a category. 'No answer' may be your most interesting finding — imputing it destroys that signal.

Whichever you choose, never silently drop rows — always report how many records the decision affected.

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DATA TRANSFORMATION

Data Manipulation Techniques

1

Recoding

Transforming values from one form to another — e.g. banding ages into groups. Decide: overwrite or new column?

2

Derived variables

A new data point computed from existing fields. Store it (fast reads) or recompute it (saves space).

3

Imputation

Substituting a missing value with an estimate — mean, median, last-known or modelled.

4

Aggregation & reduction

Summarising to a coarser grain, or sampling down. Beware introducing bias when you sample.

5

Transposing / unpivoting

Rotating rows into columns or columns into rows so the shape suits the analysis.

6

Appending & merging

Inline append discards the originals; intermediate append keeps them and creates a new combined set.

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Parsing a Messy Field with a Regular Expression

One source column holds a name, an e-mail and a phone number. You validate the pattern in RegexLab, then apply it in pandas.

```
# the raw value
'Mei Tan <mei.tan@example.sg> +65 9123 4567'

# validated in RegexLab first, then applied:
d['name']  = d.contact.str.extract(
    r'^([A-Za-z ]+?)\s*<')
d['email'] = d.contact.str.extract(
    r'([\w.%+-]+@[\w.-]+\.[A-Za-z]{2,})')
d['phone'] = d.contact.str.extract(
    r'((?:\+65 ?)?[689]\d{3} ?\d{4})')

# then normalise the phone format
d['phone'] = (d.phone.str
    .replace(r'[^0-9]', '', regex=True)
    .str[-8:])
```

^([A-Za-z]+?)\s*<

Anchored at the start; lazy match up to the first angle bracket gives the name.

[\w.%+-]+@[\w.-]+

The standard e-mail shape — local part, @, domain, then a 2+ letter TLD.

[689]\d{3} ?\d{4}

A Singapore mobile or landline: 8 digits starting 6, 8 or 9, optional +65 and spaces.

Capturing group required

pandas .str.extract returns the GROUP — without parentheses it returns NaN.

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Text, Date and Logical Functions

1

Parsing & delimiters

Split text on spaces, commas, periods, pipes or tabs — the power of a delimiter is limited by the source data.

2

Date functions

NOW() and TODAY() return the current moment; DATEDIFF() gives the interval between two dates.

3

Business-day functions

NETWORKDAYS() counts working days (but not public holidays); WEEKDAY(), WEEKNUM(), MONTH() extract parts.

4

Conditional logic

IF() tests a condition; ISNULL() substitutes a value when the expression is null.

5

AND vs OR

AND requires BOTH conditions true; OR requires EITHER to be true. Getting this backwards silently changes your filter.

6

Aggregations

SUM, COUNT, DISTINCT COUNT, AVERAGE, MIN, MAX — DISTINCT COUNT counts each value only once.

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SQL Join Types — What Each One Keeps

Join type	Keeps	Typical use
INNER JOIN	Only rows matching in BOTH tables	Confirmed transactions with a known customer
LEFT OUTER JOIN	All left rows + matches from the right	All customers, including those with no orders
RIGHT OUTER JOIN	All right rows + matches from the left	All orders, including orphans with no customer
FULL OUTER JOIN	Every row from both sides	Reconciling two systems to find what each is missing
CROSS JOIN	Every combination of both tables	Generating a complete date × product grid

WHEN IT MATTERS The silent killer: an INNER JOIN drops unmatched rows without warning. Always reconcile your record count before and after.

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QUERY PERFORMANCE

Query Optimisation

1

Filtering

WHERE restricts the result to a subset — and cuts the work the database must do.

2

Indexing

An index speeds up lookups on a column, but costs space, build time and maintenance on every write.

3

Parameterisation

Replace hard-coded values with parameters so one query serves many cases — and resists SQL injection.

4

Temporary tables

Hold an intermediate result in memory. Faster, and much easier to reason about than a giant nested query.

5

Subqueries

A query nested inside another. Readable, but often slower than the equivalent join.

6

Execution plan

The database's own visual explanation of how it will run your query — read it to find the slow step.

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Lab 4 — Explore a Dirty Dataset: Missing Values, Duplicates and Outliers

LAB 4

Perform data exploration: find missing values, duplication, redundancy or outliers (Domain 2); LO2 / K1 / A2.

You are handed a deliberately dirty sales extract and must profile it before trusting a single number. You quantify every defect class the exam names — nulls, duplicates, redundancy and outliers — and produce a data-quality report that says how bad the data is BEFORE you clean it.

YOU'LL BUILD

A data-quality profile report quantifying null counts per column, duplicate rows, and outliers detected by z-score.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas

DONE WHEN

Your profile reports 63 rows, 4 null cities, 3 null spends, 2 null order_dates and 3 duplicate rows. Two extreme outliers (99999.00 and 87500.00) are flagged at |z| > 3. The mean spend (~3348) is more than fifteen times the median (~219) — proof the outliers are distorting the mean.

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How Lab 4 Runs

1

Download this lab's dataset

step 1 of 7

2

Load it and look at the shape and dtypes fir

step 2 of 7

3

Count DUPLICATE rows and show which ones the

step 4 of 7

4

Get the descriptive summary and note how bad

step 6 of 7

5

Write the profile report to a file so the cl

step 7 of 7

YOU'LL PRODUCE

A data-quality profile report quantifying null counts per column, duplicate rows, and outliers detected by z-score.

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DOMAIN 02 · LAB 4

Lab 4 — Explore a Dirty Dataset: Missing Values, Duplicates and Outliers — Procedure 1 of 3

1

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

```
$ mkdir -p ~/dataplus/lab4 && cd ~/dataplus/lab4;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-04-explore-a-dirty-dataset-missing-values-duplicates-an;
   for f in sales_dirty.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

Load it and look at the shape and dtypes first — always know how many records you started with.

```
$ python3 -c "import pandas as pd;d=pd.read_csv('sales_dirty.csv');print(d.shape);print(d.dtypes)"
```

3

Count MISSING VALUES per column — the exam's first exploration task.

```
$ python3 -c "import pandas as pd;d=pd.read_csv('sales_dirty.csv');print(d.isnull().sum())"
```


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Lab 4 — Explore a Dirty Dataset: Missing Values, Duplicates and Outliers — Procedure 2 of 3

4

Count DUPLICATE rows and show which ones they are.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('sales_dirty.csv');
   print('dupes:',d.duplicated().sum());
   print(d[d.duplicated(keep=False)])"
```

5

Detect OUTLIERS with a z-score — any |z| above 3 is the standard flag.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('sales_dirty.csv');
   s=pd.to_numeric(d['spend'],errors='coerce');
   z=(s-s.mean())/s.std();
   print(d.loc[z[abs(z)>3].index])"
```

6

Get the descriptive summary and note how badly the 99999 distorts the mean versus the median.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('sales_dirty.csv');
   print(d['spend'].describe());
   print('median',d['spend'].median())"
```


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Lab 4 — Explore a Dirty Dataset: Missing Values, Duplicates and Outliers — Procedure 3 of 3

Write the profile report to a file so the cleaning lab can be measured against it.

7

```
$ python3 -c "import pandas as
   pd;d=pd.read_csv('sales_dirty.csv');open('profile.txt','w').write(str(d.isnull().sum())+'\ndupes:
   '+str(d.duplicated().sum())+'\nmean: '+str(d.spend.mean())+'\nmedian: '+str(d.spend.median()))" && cat
   profile.txt
```


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Lab 4 — Verify Your Work

✅ Expected result

Your profile reports 63 rows, 4 null cities, 3 null spends, 2 null order_dates and 3 duplicate rows. Two extreme outliers (99999.00 and 87500.00) are flagged at |z| > 3. The mean spend (~3348) is more than fifteen times the median (~219) — proof the outliers are distorting the mean.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

ModuleNotFoundError: pandas

Run pip3 install pandas. On Killercoda add --break-system-packages if pip refuses.

The download produced an empty file

Check your internet connection and the BASE URL, or copy sales_dirty.csv from the repo folder you cloned.

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Lab 5 — Build Parsing Patterns in RegexLab and Apply Them

LAB 5

Apply data transformation: cleansing, parsing and formatting data (Domain 2); LO2 / K1.

Real source fields arrive as one messy string — 'Mei Tan <mei.tan@example.sg> +65 9123 4567'. You use RegexLab to build and validate the extraction patterns interactively, then apply the proven patterns in pandas to split one dirty column into three clean, typed fields.

YOU'LL BUILD

Three validated regex patterns (name, email, phone) and a cleaned CSV with the single contact column parsed into three columns.

TOOLCHAIN

RegexLab (https://alfredang.github.io/regexgenerator/), Killercoda Ubuntu, Python 3, pandas

DONE WHEN

contacts_clean.csv holds 30 rows, each with a name, an email and a normalised 8-digit phone. All 30 rows match all three patterns — no NaN in any column, no angle brackets, no +65 prefixes and no leftover spaces.

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How Lab 5 Runs

1. **Download this lab's dataset** — step 1 of 9
2. **Build the EMAIL pattern and watch the match** — step 3 of 9
3. **Build the NAME pattern** — step 5 of 9
4. **Apply the SAME patterns you validated in Reg** — step 7 of 9
5. **Save the cleaned, parsed output** — step 9 of 9

YOU'LL PRODUCE

Three validated regex patterns (name, email, phone) and a cleaned CSV with the single contact column parsed into three columns.

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Lab 5 — Build Parsing Patterns in RegexLab and Apply Them — Procedure 1 of 3

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

1

```
$ mkdir -p ~/dataplus/lab5 && cd ~/dataplus/lab5;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-05-build-parsing-patterns-in-regexlab-and-apply-them;
   for f in contacts.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

Paste these three messy contact records into the Test String box: Mei Tan <mei.tan@example.sg> +65 9123 4567

Build the EMAIL pattern and watch the match count update live. Confirm it matches all three records.

3

```
$ [a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}
```


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Lab 5 — Build Parsing Patterns in RegexLab and Apply Them — Procedure 2 of 3

4 **Build the SINGAPORE PHONE pattern — 8 digits starting 6, 8 or 9, with optional +65 and spaces.**

```
$ (?:\+65[ ]?)?[689][0-9]{3}[ ]?[0-9]{4}
```

5 **Build the NAME pattern — everything before the first angle bracket, trimmed.**

```
$ ^([A-Za-z ]+?)\s*<
```

6 Use the Substitution panel to confirm your pattern replaces cleanly before you trust it in code.

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Lab 5 — Build Parsing Patterns in RegexLab and Apply Them — Procedure 3 of 3

7 Apply the SAME patterns you validated in RegexLab, using pandas str.extract.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('contacts.csv');
   d['name']=d.contact.str.extract(r'^([A-Za-z ]+?)\s*<');
   d['email']=d.contact.str.extract(r'([a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,})');
   d['phone']=d.contact.str.extract(r'((?:\+65 ?)?[689][0-9]{3} ?[0-9]{4})');
   print(d[['id','name','email','phone']])"
```

8 Normalise the phone format — strip +65 and spaces so every value has the same shape.

```
$ python3 -c "import pandas as pd,re;
   d=pd.read_csv('contacts.csv');
   d['phone']=d.contact.str.extract(r'((?:\+65 ?)?[689][0-9]{3} ?[0-9]{4})')[0].str.replace(r'[^0-9]','',regex=True).str[-8:];
   print(d[['id','phone']])"
```

9 Save the cleaned, parsed output.

```
$ python3 -c "import pandas as pd;d=pd.read_csv('contacts.csv');d['name']=d.contact.str.extract(r'^([A-Za-z
   ]+?)\s*<');d['email']=d.contact.str.extract(r'([a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,})
   ');d['phone']=d.contact.str.extract(r'((?:\+65 ?)?[689][0-9]{3}
   ?[0-9]{4})')[0].str.replace(r'[^0-9]','',regex=True).str[-8:];d[['id','name','email','phone']].to_csv('contact
   s_clean.csv',index=False)" && cat contacts_clean.csv
```


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Lab 5 — Verify Your Work

✅ Expected result

contacts_clean.csv holds 30 rows, each with a name, an email and a normalised 8-digit phone. All 30 rows match all three patterns — no NaN in any column, no angle brackets, no +65 prefixes and no leftover spaces.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

RegexLab shows 0 matches

Check the flags — you usually want 'g' (global) so every record is matched, not just the first.

extract returns NaN

pandas needs a capturing group. Confirm your pattern has parentheses around the part you want.

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Lab 6 — Derive Structured Fields from Raw Address Data (IP Calculator)

LAB 6

Apply data transformation: derived variables and formatting; use data acquisition methods (Domain 2); LO2 / A2.

A derived variable is a new field computed from existing data — one of the exam's named manipulation techniques. You take raw CIDR address data, use the IP Calculator to derive the network fields by hand first, then reproduce the same derivation in code and prove the two agree.

YOU'LL BUILD

A dataset enriched with four derived columns (network address, broadcast, usable hosts, subnet class) verified against the calculator.

TOOLCHAIN

IP Calculator (https://alfredang.github.io/ipcalculator/), Killercoda Ubuntu, Python 3, ipaddress module

DONE WHEN

The code and the IP Calculator agree: /24 gives 254 usable hosts, /22 gives 1022, and /29 gives 6. sites_enriched.csv carries the derived columns alongside the original CIDR.

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How Lab 6 Runs

1. **Download this lab's dataset**
   step 1 of 7
2. **Enter 192**
   step 2 of 7
3. **Derive the same four fields in code**
   step 4 of 7
4. **Decide the storage trade-off the exam asks a**
   step 6 of 7
5. **Save the enriched dataset**
   step 7 of 7

YOU'LL PRODUCE

A dataset enriched with four derived columns (network address, broadcast, usable hosts, subnet class) verified against the calculator.

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DOMAIN 02 · LAB 6

Lab 6 — Derive Structured Fields from Raw Address Data (IP Calculator) — Procedure 1 of 3

1. **Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.**

```
$ mkdir -p ~/dataplus/lab6 && cd ~/dataplus/lab6;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-06-derive-structured-fields-from-raw-address-data-ip-ca;
   for f in sites.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2. Enter 192.168.10.0/24 and record the derived values: network, broadcast, usable host count and mask.

3. Repeat for 10.0.5.0/22 and 172.16.8.0/29 — note how the usable-host count changes with the prefix.

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Lab 6 — Derive Structured Fields from Raw Address Data (IP Calculator) — Procedure 2 of 3

4 **Derive the same four fields in code — this is the DERIVED VARIABLE technique from the exam objectives.**

```
$ python3 -c "import pandas as pd,ipaddress as
   ip;d=pd.read_csv('sites.csv');n=d.cidr.map(ip.ip_network);d['network']=[str(x.network_address) for x in
   n];d['broadcast']=[str(x.broadcast_address) for x in n];d['usable_hosts']=[x.num_addresses-2 for x in
   n];d['mask']=[str(x.netmask) for x in n];print(d)"
```

5 Compare every derived value against what the IP Calculator gave you — they must match exactly.

6 Decide the storage trade-off the exam asks about: store the derived columns (fast reads, more space) or recompute on demand (less space, slower). Write your choice and the reason.

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Lab 6 — Derive Structured Fields from Raw Address Data (IP Calculator) — Procedure 3 of 3

Save the enriched dataset.

7

```
$ python3 -c "import pandas as pd,ipaddress as
  ip;d=pd.read_csv('sites.csv');n=d.cidr.map(ip.ip_network);d['network']=[str(x.network_address) for x in
  n];d['usable_hosts']=[x.num_addresses-2 for x in n];d.to_csv('sites_enriched.csv',index=False)" && cat
  sites_enriched.csv
```


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Lab 6 — Verify Your Work

✅ Expected result

The code and the IP Calculator agree: /24 gives 254 usable hosts, /22 gives 1022, and /29 gives 6. sites_enriched.csv carries the derived columns alongside the original CIDR.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

ValueError: has host bits set

The address is not a valid network address for that prefix. Use ip_network(x, strict=False) or correct the CIDR.

usable_hosts is negative for /31 or /32

Those prefixes have no usable host range by the -2 convention. Note this edge case in your report.

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Lab 7 — Integrate Multiple Datasets with SQL Joins and Cleanse the Result

LAB 7

Use data acquisition methods: data integration and queries to gather and combine data (Domain 2); LO2 / A1 / A2.

This is the integration lab that LO1 and LO2 both rest on. You load three separate source extracts into SQLite, combine them with the four join types, watch how inner versus left join silently changes your record count, and then cleanse the merged result.

YOU'LL BUILD

A single integrated, deduplicated analysis table built from three sources, with the record count reconciled at every join.

TOOLCHAIN

Killercoda Ubuntu, SQLite 3, SQL joins, Python 3/pandas

DONE WHEN

The sources hold 31 customers and 81 orders. The INNER JOIN returns only 80 rows — order 999 is dropped because customer 77 has no master record. Four customers have no orders at all and survive only via the LEFT JOIN. The regional table reports every region's revenue against its target.

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How Lab 7 Runs

1

Download this lab's dataset

step 1 of 8

2

INNER JOIN

step 3 of 8

3

LEFT JOIN from customers

step 5 of 8

4

Add the derived performance measure and pers

step 7 of 8

5

Reconcile: explain in one line why the inner

step 8 of 8

YOU'LL PRODUCE

A single integrated, deduplicated analysis table built from three sources, with the record count reconciled at every join.

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Lab 7 — Integrate Multiple Datasets with SQL Joins and Cleanse the Result — Procedure 1 of 3

1

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

```
$ mkdir -p ~/dataplus/lab7 && cd ~/dataplus/lab7;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-07-integrate-multiple-datasets-with-sql-joins-and-clean;
   for f in customers.csv orders.csv targets.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

Load all three into SQLite in one go.

```
$ sqlite3 integrate.db <<'EOF'
```

3

INNER JOIN — returns only matched rows. Count them and note what you silently lost.

```
$ sqlite3 -header -column integrate.db "SELECT COUNT(*) AS inner_rows FROM orders o JOIN customers c ON
   c.customer_id=o.customer_id;"
```


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Lab 7 — Integrate Multiple Datasets with SQL Joins and Cleanse the Result — Procedure 2 of 3

4 LEFT JOIN from orders — keeps order 104 whose customer is missing, showing NULL in the master fields.

```
$ sqlite3 -header -column integrate.db "SELECT o.order_id, o.customer_id, c.name FROM orders o LEFT JOIN
   customers c ON c.customer_id=o.customer_id;"
```

5 LEFT JOIN from customers — keeps John Lee, who has no orders at all.

```
$ sqlite3 -header -column integrate.db "SELECT c.name, o.order_id FROM customers c LEFT JOIN orders o ON
   c.customer_id=o.customer_id;"
```

6 Build the integrated analysis table joining all three sources and aggregating per region.

```
$ sqlite3 -header -column integrate.db "SELECT c.region, COUNT(o.order_id) AS orders, ROUND(SUM(o.amount),2) AS
   revenue, t.target FROM customers c LEFT JOIN orders o ON c.customer_id=o.customer_id JOIN targets t ON
   t.region=c.region GROUP BY c.region, t.target;"
```


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Lab 7 — Integrate Multiple Datasets with SQL Joins and Cleanse the Result — Procedure 3 of 3

Add the derived performance measure and persist the result.

7

```
$ sqlite3 -header -column integrate.db "CREATE TABLE regional AS SELECT c.region, COUNT(o.order_id) AS orders,
   COALESCE(SUM(o.amount),0) AS revenue, t.target, ROUND(COALESCE(SUM(o.amount),0)*100.0/t.target,1) AS
   pct_of_target FROM customers c LEFT JOIN orders o ON c.customer_id=o.customer_id JOIN targets t ON
   t.region=c.region GROUP BY c.region,t.target; SELECT * FROM regional;"
```

8

Reconcile: explain in one line why the inner join returned 4 rows but there are 5 orders.

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Lab 7 — Verify Your Work

✅ Expected result

The sources hold 31 customers and 81 orders. The INNER JOIN returns only 80 rows — order 999 is dropped because customer 77 has no master record. Four customers have no orders at all and survive only via the LEFT JOIN. The regional table reports every region's revenue against its target.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

'.import' left the header as a data row

Older SQLite lacks --skip 1. Delete it after import: DELETE FROM customers WHERE customer_id='customer_id';

SUM returns NULL for East

That is correct SQL — no rows to sum. COALESCE(...,0) is what turns it into a reportable zero.

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DOMAIN 02 RECAP

Recap — Data Acquisition and Preparation

1. **ETL vs ELT**
   ETL transforms before loading; ELT loads raw and transforms later — choose by how stable your requirements are.
2. **Data sources**
   Data comes from databases, APIs, scraping, files, logs and public repositories, each with its own reliability risk.
3. **Profiling first**
   Profiling is the disciplined first pass — quantify the defects before you clean, so you can measure the cleaning.
4. **Defect classes**
   The defect classes are missing values, duplicates, redundancy, invalid data and outliers.
5. **Manipulation**
   Manipulation covers recoding, derived variables, imputation, aggregation, transposing, appending and parsing.
6. **Joins**
   Joins combine datasets — and an INNER JOIN silently drops unmatched rows, so always reconcile record counts.
7. **Query performance**
   Indexes, filters, parameters, temp tables and the execution plan are how you make a slow query fast.

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Tea Break

15 minutes

03

DOMAIN 03 · 24% OF THE EXAM

Data Analysis

Descriptive statistics · inferential methods · analysis types · communicating results · troubleshooting

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THE HEAVIEST DOMAIN ON THE EXAM

Domain 3 — Key Concepts

1. **Descriptive statistics**
   Descriptive statistics summarise a dataset: central tendency (mean, median, mode) and dispersion (range, variance, standard deviation).
2. **Mean vs median**
   The mean is pulled by outliers; the median resists them — which one you report changes the story the data tells.
3. **Z-scores find outliers**
   Z-scores express how many standard deviations a value sits from the mean, and are the standard outlier test.
4. **The empirical rule**
   The empirical rule says ~99.7% of normally distributed data falls within three standard deviations of the mean.
5. **Inferential statistics**
   Inferential statistics generalise beyond the sample: t-tests, p-values, chi-square, correlation and regression.
6. **p-values and causation**
   A p-value below 0.05 indicates the observed difference is unlikely to be due to chance; correlation still never proves causation.
7. **Choosing the analysis type**
   Analysis types — exploratory, performance, trend, gap and link analysis — each answer a different business question.

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Choosing Your Analysis Type

1

Exploratory analysis

The first look at a new dataset — what cleaning, profiling and transformation does it need?

2

Performance analysis

Measures a product, outcome or scenario against a defined objective, using realistic KPIs.

3

Trend analysis

Uses historical data to project a future outcome. Past performance never guarantees future results.

4

Gap analysis

The difference (the delta) between the present state and the desired future state.

5

Link analysis

How one data point connects to others — a network of nodes joined by links.

6

Hypothesis-driven

Start with a testable statement, then gather evidence to support or reject it.

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Central Tendency — Three Different 'Averages'

1

Mean

The sum divided by the count. Uses every value — and is therefore dragged by outliers.

2

Median

The middle value when sorted. Resistant to outliers, which is why salary and house prices use it.

3

Mode

The most frequently occurring value. The only average that works on categorical data.

4

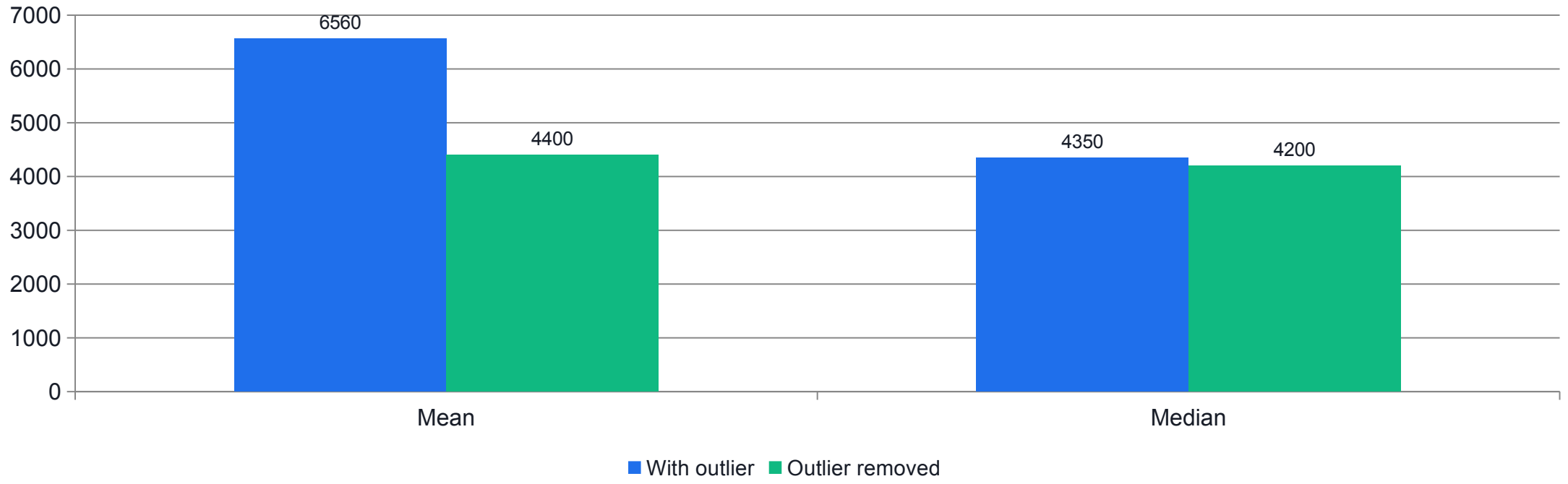
Which to report

Symmetric data → mean. Skewed data or outliers present → median. Categories → mode.

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WHY THE CHOICE MATTERS

The Outlier Effect — Mean vs Median (Lab 8 data)



WHAT THE DATA SHOWS

One executive salary moves the mean by over 2,100 but the median by only 150. When a distribution is skewed, reporting the mean overstates what a typical person earns — which is exactly what you prove in Lab 8.

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Dispersion — How Spread Out Is the Data?

1

Minimum / Maximum

The smallest and largest values in the dataset.

2

Range

Maximum minus minimum. A large range is an early clue that outliers are present.

3

Variance (σ^2)

The average of the squared differences from the mean. Its units are squared, so it is hard to interpret directly.

4

Standard deviation (σ)

The square root of the variance — back in the original units, so you can actually read it.

5

Z-score

How many standard deviations a value sits from the mean: $z = (x - \bar{x}) / s$.

6

Outlier rule of thumb

$|z| > 3$ is a strong outlier flag; $|z| > 2$ is worth investigating.

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DISTRIBUTION

Distribution and Frequency

1

Normal distribution

The symmetric bell curve — the empirical rule says ~99.74% of data lies within 3 standard deviations.

2

Parametric data

Fits a known distribution (usually normal), so mean-and-SD methods are valid.

3

Nonparametric data

Does not fit a normal distribution — use distribution-independent methods and the median.

4

Skew

An asymmetric distribution with a long tail on one side, which pulls the mean away from the median.

5

Frequency

How many times a value appears; frequency percent expresses it relative to the whole dataset.

6

Histogram

The chart that shows a distribution — choose your bin count carefully and include the outliers.

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INFERENTIAL STATISTICAL METHODS

Inferential statistics let you speak beyond your sample.

Descriptive statistics summarise the data you have. Inferential statistics — t-tests, chi-square, regression — let you draw conclusions about a whole population from a sample, and state how confident you are in that conclusion.

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Hypothesis Testing, Step by Step



THE DISCIPLINE

State the hypotheses BEFORE you look at the result. Choosing the hypothesis to fit the answer you wanted is the most common analytical malpractice.

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Null vs Alternative Hypothesis, and the Two Errors

Concept	Meaning	In plain language
Null hypothesis (H0)	No relationship between the variables	'The new page makes no difference'
Alternative (H1)	A relationship exists	'The new page lifts order value'
p-value	Probability the difference arose by chance	Below 0.05 → unlikely to be chance
Reject H0	The evidence supports a real effect	'We are confident the page works'
Type I error	Rejecting a TRUE null hypothesis	False alarm — you ship a change that does nothing
Type II error	Failing to reject a FALSE null	Missed opportunity — you bin a change that worked

WHEN IT MATTERS A p-value below 0.05 means the result is statistically significant — it does NOT tell you the effect is large or commercially important.

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STATISTICAL METHODS

The Inferential Tests You Must Know

1

T-test

Compares the MEANS of two groups to see whether the difference is significant. Used in every A/B test.

2

Chi-square

Compares observed counts against expected counts for CATEGORICAL data.

3

Chi-square: independence

Tests whether two categorical variables are related to each other.

4

Chi-square: goodness of fit

Tests whether observed counts match an expected baseline distribution.

5

Regression analysis

Estimates the relationship between a dependent variable and one or more independent variables.

6

Confidence interval

The range within which the true population value is likely to fall, at a stated confidence level.

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REGRESSION ANALYSIS

Reading a Regression Output

Lab 10 fits revenue against marketing spend. Here is what each number in the output actually tells you.

```
from scipy import stats

lr = stats.linregress(d.spend, d.revenue)

# slope      =  6.598
# intercept  = 52.114
# rvalue     =  0.9992
# pvalue     =  1.4e-12

# the fitted model:
#   revenue = 6.598 * spend + 52.114

# predict at a new spend level
lr.slope * 40 + lr.intercept
#  -> 316.0
```

Slope = 6.598

Every extra 1 unit of spend is associated with 6.6 more units of revenue.

Intercept = 52.1

The modelled revenue at zero spend — often not meaningful in the real world.

R^2 = 0.998

99.8% of the variation in revenue is explained by spend. Very high — and suspiciously so.

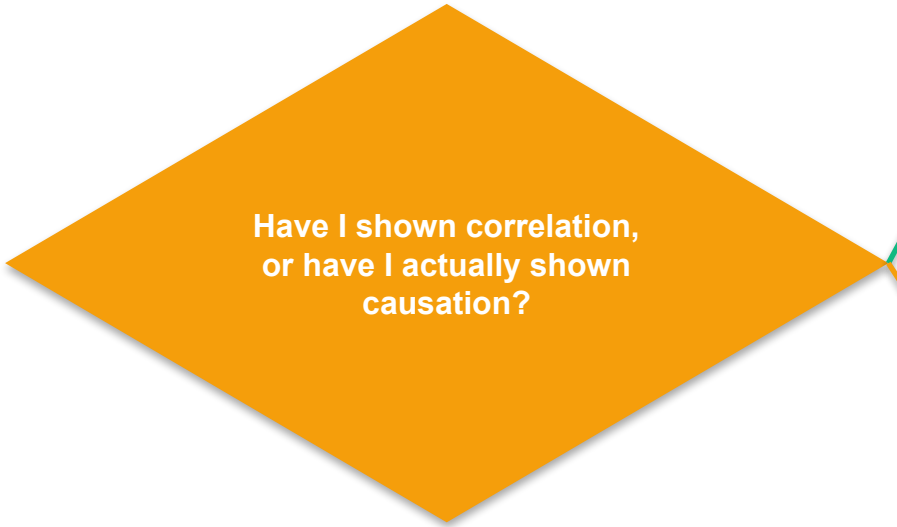
p = 1.4e-12

Far below 0.05, so the relationship is statistically significant, not chance.

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THE ANALYST'S DUTY

Two Variables Move Together. Now What?



CORRELATION — what you have

Pearson's r measures the strength and direction of a linear relationship. R^2 is the proportion of variance explained. This is a pattern, nothing more.

CAUSATION — what you need

A controlled experiment, a plausible mechanism, correct time ordering, and the ruling out of confounding variables. Observational data alone can never prove it.

In Lab 10 staff numbers correlate with revenue at $r \approx 0.99$ — but growth over time drives both. Correlation is not causation.

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Troubleshooting Your Analysis

1

The numbers don't reconcile

Check the record count at every join and filter. An INNER JOIN or a WHERE clause has probably dropped rows.

2

The mean looks wrong

Check for outliers and nulls. pandas skips NaN in a mean — so your denominator may not be what you think.

3

Everything correlates

You may be measuring the same thing twice, or both variables are driven by time. Plot them before you trust r.

4

The result is too clean

Real business data is noisy. A perfect fit usually means leakage, a synthetic dataset, or a duplicated column.

5

Dates behave strangely

Confirm the parse format and the timezone. A DD/MM vs MM/DD mix-up silently corrupts every time series.

6

Nobody trusts the number

Show the lineage: the source, the filters applied, the row counts and the date the data was refreshed.

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Communicating Results to Different Audiences

1

C-level executives

The big picture only. Lead with the decision and the number that supports it — never the methodology.

2

Management

Performance against target, by unit, with the exceptions highlighted and an action attached.

3

Technical experts

They will ask about your method, your sample and your assumptions. Bring the detail and the caveats.

4

External stakeholders

Know exactly what may and may not be shared. Aggregate and de-identify before it leaves the building.

5

General public

Plain language, no jargon, one message per visual, and a clearly stated source.

6

The universal rule

Lead with the answer, then the evidence. Never make your audience assemble the conclusion themselves.

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Lab 8 — Descriptive Statistics and the Outlier That Moves the Mean

LAB 8

Select statistical methods: apply basic statistical techniques to data (Domain 3); LO3 / K2 / A3.

You compute every descriptive statistic the exam names — mean, median, mode, range, variance, standard deviation and z-score — on a real salary dataset, then remove one outlier and watch which statistics move and which do not. This is the exam's central-tendency-versus-robustness point, proven with numbers.

YOU'LL BUILD

A descriptive-statistics report showing mean, median, mode, range, variance, SD and z-scores, computed with and without the outlier.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas, statistics

DONE WHEN

Across 61 employees the mean is ~4978 but the median is only 4550. Removing the single Executive salary drops the mean to ~4628 while the median barely moves (4550 → 4525). E999 is flagged at z ≈ 7.5, far beyond the |z| > 3 threshold. Your report recommends the MEDIAN as the typical salary.

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How Lab 8 Runs

1. **Download this lab's dataset**
   step 1 of 8
2. **Compute DISPERSION**
   step 3 of 8
3. **Now remove the outlier and recompute the SAM**
   step 5 of 8
4. **Compute the departmental summary**
   step 7 of 8
5. **Write your recommendation: which single numb**
   step 8 of 8

YOU'LL PRODUCE

A descriptive-statistics report showing mean, median, mode, range, variance, SD and z-scores, computed with and without the outlier.

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Lab 8 — Descriptive Statistics and the Outlier That Moves the Mean — Procedure 1 of 3

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

1

```
$ mkdir -p ~/dataplus/lab8 && cd ~/dataplus/lab8;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-08-descriptive-statistics-and-the-outlier-that-moves-th;
   for f in salaries.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

Compute CENTRAL TENDENCY — mean, median and mode together.

2

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('salaries.csv');
   s=d.salary;
   print('mean',round(s.mean(),2));
   print('median',s.median());
   print('mode',s.mode().tolist())"
```

Compute DISPERSION — min, max, range, variance and standard deviation.

3

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('salaries.csv');
   s=d.salary;
   print('min',s.min(),'max',s.max(),'range',s.max()-s.min());
   print('variance',round(s.var(),2),'sd',round(s.std(),2))"
```


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Lab 8 — Descriptive Statistics and the Outlier That Moves the Mean — Procedure 2 of 3

4 Compute the Z-SCORE for every row and flag anything beyond the standard |z| > 3 threshold.

```
$ python3 -c "import pandas as
   pd;d=pd.read_csv('salaries.csv');s=d.salary;d['z']=((s-s.mean())/s.std()).round(2);print(d);print('FLAGGED:');
   print(d[abs(d.z)>3])"
```

5 Now remove the outlier and recompute the SAME statistics.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('salaries.csv');
   c=d[d.salary<20000].salary;
   print('mean',round(c.mean(),2));
   print('median',c.median());
   print('sd',round(c.std(),2))"
```

6 Compare the two runs side by side and record which statistic moved most.

```
$ python3 -c "import pandas as
   pd;d=pd.read_csv('salaries.csv');a=d.salary;b=d[d.salary<20000].salary;print(f'mean {a.mean():>9.2f} ->
   {b.mean():>8.2f}');print(f'median {a.median():>9.2f} -> {b.median():>8.2f}');print(f'sd {a.std():>9.2f} ->
   {b.std():>8.2f}')"
```


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Lab 8 — Descriptive Statistics and the Outlier That Moves the Mean — Procedure 3 of 3

Compute the departmental summary — this is the aggregation a manager actually asks for.

7

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('salaries.csv');
   print(d.groupby('dept').salary.agg(['count','mean','median','std']).round(2))"
```

8

Write your recommendation: which single number should be reported to the board, and why.

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Lab 8 — Verify Your Work

✅ Expected result

Across 61 employees the mean is ~4978 but the median is only 4550. Removing the single Executive salary drops the mean to ~4628 while the median barely moves (4550 → 4525). E999 is flagged at z ≈ 7.5, far beyond the |z| > 3 threshold. Your report recommends the MEDIAN as the typical salary.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

mode returns several values

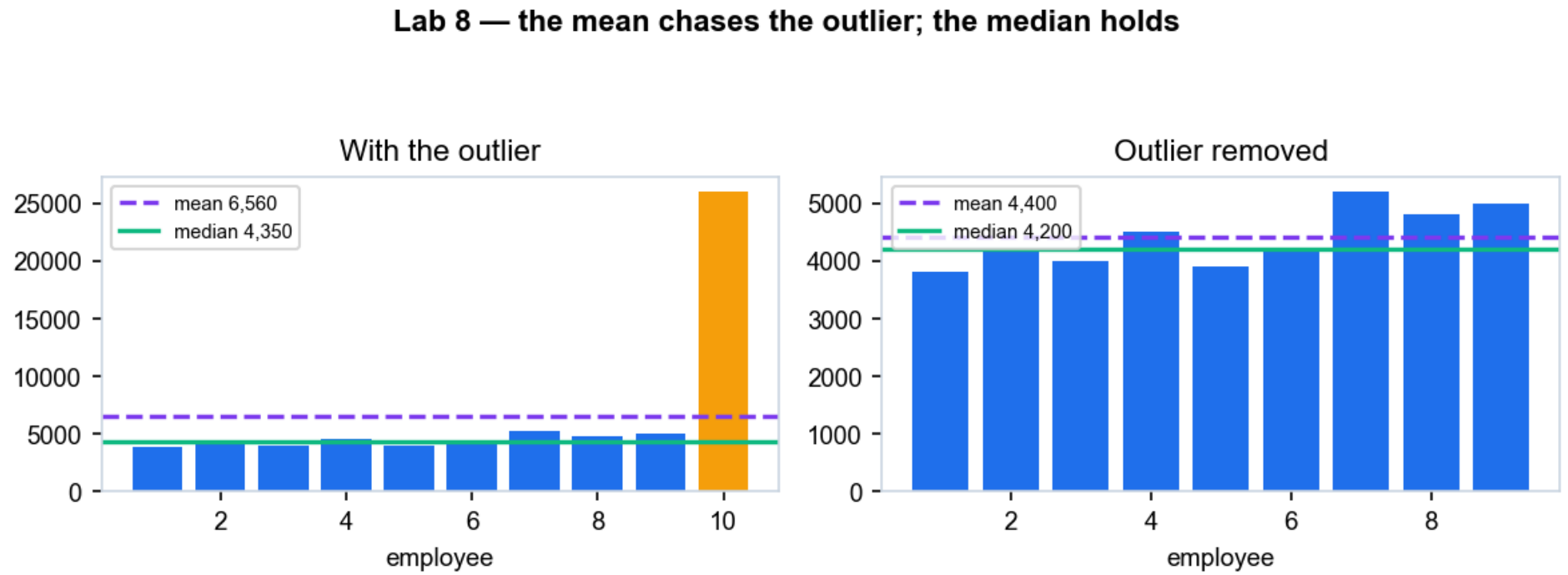
A dataset with no repeated value returns every value. pandas .mode() correctly returns a list — report it as 'no single mode'.

Variance looks enormous

Variance is in squared units. Report the standard deviation (its square root) instead, which is in dollars.

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Lab 8 — Expected Output



WHAT YOU SHOULD SEE

This is what your own run should produce. Across 61 employees the mean is ~4978 but the median is only 4550. Removing the single Executive salary drops the mean to ~4628 while the median barely moves (4550 → 4525).

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Lab 9 — Hypothesis Testing: t-test, p-value and the Two Error Types

LAB 9

Select statistical methods: inferential techniques, hypothesis testing (Domain 3); LO3 / K2 / A3.

A marketing team claims the new checkout page lifts order value. You state the null and alternative hypotheses, run a two-sample t-test, read the p-value against the 0.05 threshold, and state your conclusion in business language — including which error type you would be risking if you are wrong.

YOU'LL BUILD

A completed hypothesis test: stated H0/H1, computed t-statistic and p-value, an accept/reject decision, and the business recommendation.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas, scipy

DONE WHEN

With 60 observations per group the t-test returns t ≈ -11.3 and p ≈ 1.9e-20, far below 0.05, so you REJECT H0. Group B averages about 49.65 against 41.14 for group A — a lift of roughly 8.5, with a 95% confidence interval that excludes zero.

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How Lab 9 Runs

1. **Download this lab's dataset**
   step 1 of 8
2. **Look at the group means first**
   step 3 of 8
3. **Apply the 0**
   step 5 of 8
4. **Now the error-type question**
   step 7 of 8
5. **Write the one-paragraph recommendation a man**
   step 8 of 8

YOU'LL PRODUCE

A completed hypothesis test: stated H0/H1, computed t-statistic and p-value, an accept/reject decision, and the business recommendation.

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Lab 9 — Hypothesis Testing: t-test, p-value and the Two Error Types — Procedure 1 of 3

1

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

```
$ mkdir -p ~/dataplus/lab9 && cd ~/dataplus/lab9;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-09-hypothesis-testing-t-test-p-value-and-the-two-error;
   for f in abtest.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

STATE THE HYPOTHESES before you look at any result — this is the discipline the exam tests. H0: there is no difference in mean order value. H1: the new page has a higher mean order value.

3

Look at the group means first — a difference here is necessary but NOT sufficient.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('abtest.csv');
   print(d.groupby('group').order_value.agg(['count','mean','std']).round(2))"
```


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Lab 9 — Hypothesis Testing: t-test, p-value and the Two Error Types — Procedure 2 of 3

4 Run the two-sample t-test and read the p-value.

```
$ python3 -c "import pandas as pd;from scipy import
   stats;d=pd.read_csv('abtest.csv');a=d[d.group=='A'].order_value;b=d[d.group=='B'].order_value;t,
   p=stats.ttest_ind(a,b);print('t =',round(t,4));print('p =',round(p,6))"
```

5 Apply the 0.05 decision rule explicitly.

```
$ python3 -c "import pandas as pd;from scipy import
   stats;d=pd.read_csv('abtest.csv');a=d[d.group=='A'].order_value;b=d[d.group=='B'].order_value;t,
   p=stats.ttest_ind(a,b);print('REJECT H0 - the difference is statistically significant' if p<0.05 else 'FAIL TO
   REJECT H0')"
```

6 Compute the 95% confidence interval for the difference so you can report a range, not just a verdict.

```
$ python3 -c "import pandas as pd,numpy as np;from scipy import
   stats;d=pd.read_csv('abtest.csv');a=d[d.group=='A'].order_value;b=d[d.group=='B'].order_value;diff=b.mean()
   -a.mean();se=np.sqrt(a.var()/len(a)+b.var()/len(b));print('diff',round(diff,2),'95%
   CI',(round(diff-1.96*se,2),round(diff+1.96*se,2)))"
```


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Lab 9 — Hypothesis Testing: t-test, p-value and the Two Error Types — Procedure 3 of 3

7 Now the error-type question. Write down: if you reject H0 and you are WRONG, which error is that (Type I) and what does it cost the business? If you fail to reject and you are wrong (Type II), what does that cost?

8 Write the one-paragraph recommendation a manager could act on — no statistics jargon.

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Lab 9 — Verify Your Work

✅ Expected result

With 60 observations per group the t-test returns t ≈ -11.3 and p ≈ 1.9e-20, far below 0.05, so you REJECT H0. Group B averages about 49.65 against 41.14 for group A — a lift of roughly 8.5, with a 95% confidence interval that excludes zero.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

ModuleNotFoundError: scipy

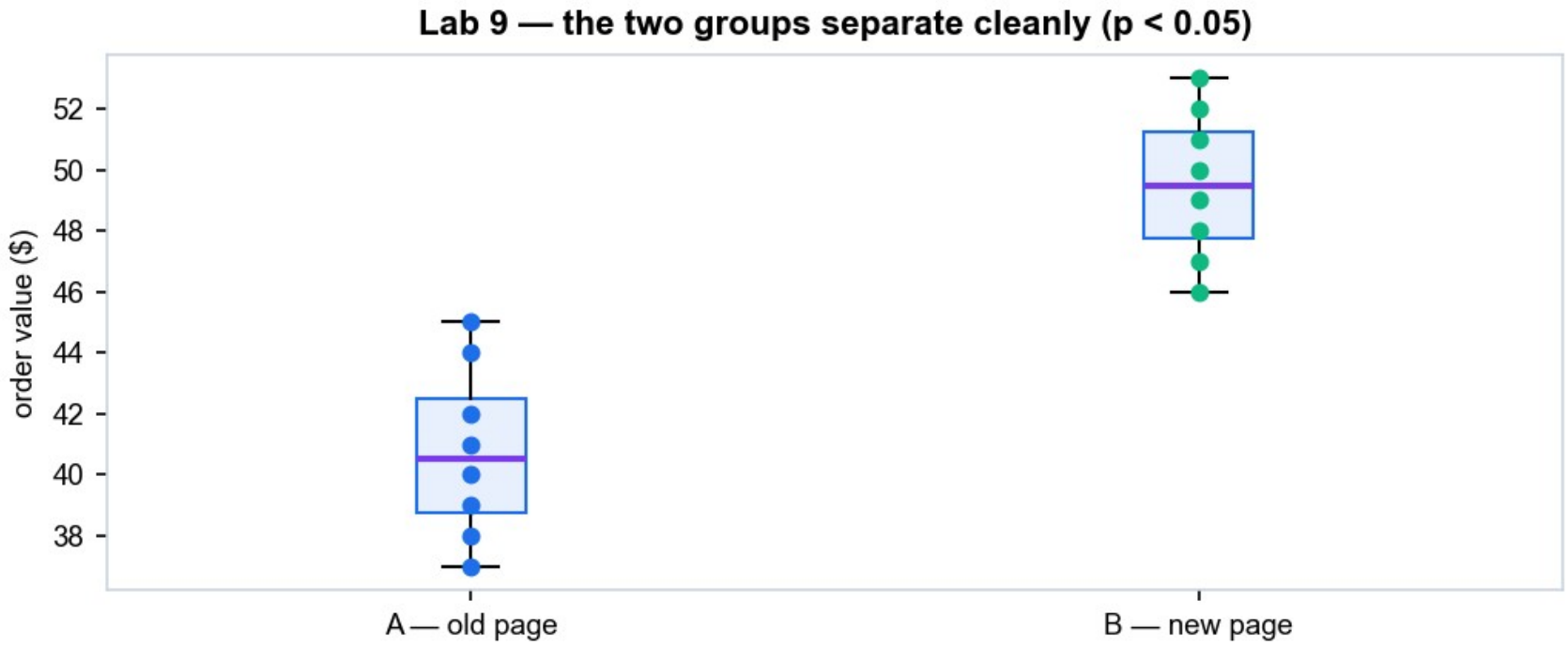
Run pip3 install scipy, adding --break-system-packages if Killercoda's pip refuses.

p-value is nan

One group has fewer than two values or zero variance. Check your CSV loaded all 16 rows with d.shape.

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Lab 9 — Expected Output



WHAT YOU SHOULD SEE

This is what your own run should produce. With 60 observations per group the t-test returns t ≈ -11.3 and p ≈ 1.9e-20, far below 0.05, so you REJECT H0.

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Lab 10 — Correlation, Regression and the Causation Trap

LAB 10

Select statistical methods: correlation and regression; troubleshoot analysis issues (Domain 3); LO3 / LO5 / A3 / A4.

You measure the relationship between marketing spend and revenue with Pearson's r and R-squared, fit a regression line, use it to predict — and then meet a third dataset where two variables correlate almost perfectly with no causal link at all. Recognising that trap is an exam objective and a professional duty.

YOU'LL BUILD

A correlation matrix, a fitted regression equation with R-squared, a prediction, and a written spurious-correlation analysis.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas, scipy

DONE WHEN

Across 36 months spend and revenue correlate at r ≈ 0.991 with R-squared ≈ 0.982, giving revenue ≈ 6.55 × spend + 54.1. Staff headcount ALSO correlates with revenue at r ≈ 0.970 — but growth over time drives both, so that second relationship is not causal.

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How Lab 10 Runs

1

Download this lab's dataset

step 1 of 8

2

Get Pearson's r and its p-value for spend ve

step 3 of 8

3

Use the model to PREDICT revenue at a spend

step 5 of 8

4

THE TRAP

step 7 of 8

5

Explain in writing: does hiring staff CAUSE

step 8 of 8

YOU'LL PRODUCE

A correlation matrix, a fitted regression equation with R-squared, a prediction, and a written spurious-correlation analysis.

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Lab 10 — Correlation, Regression and the Causation Trap — Procedure 1 of 3

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

1

```
$ mkdir -p ~/dataplus/lab10 && cd ~/dataplus/lab10;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-10-correlation-regression-and-the-causation-trap;
   for f in marketing.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

Compute the full CORRELATION MATRIX — every pair at once.

2

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('marketing.csv');
   print(d[['spend','revenue','staff']].corr().round(4))"
```

Get Pearson's r and its p-value for spend versus revenue specifically.

3

```
$ python3 -c "import pandas as pd;from scipy import
   stats;d=pd.read_csv('marketing.csv');r,p=stats.pearsonr(d.spend,d.revenue);print('r
   =',round(r,4));print('r-squared =',round(r**2,4));print('p =',round(p,8))"
```


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Lab 10 — Correlation, Regression and the Causation Trap — Procedure 2 of 3

4 **Fit the REGRESSION LINE and read off the slope and intercept.**

```
$ python3 -c "import pandas as pd;
   from scipy import stats;
   d=pd.read_csv('marketing.csv');
   lr=stats.linregress(d.spend,d.revenue);
   print(f'revenue = {lr.slope:.3f} * spend + {lr.intercept:.3f}');
   print('R-squared =',round(lr.rvalue**2,4))"
```

5 **Use the model to PREDICT revenue at a spend level you have never observed.**

```
$ python3 -c "import pandas as pd;
   from scipy import stats;
   d=pd.read_csv('marketing.csv');
   lr=stats.linregress(d.spend,d.revenue);
   print('predicted revenue at spend=40:',round(lr.slope*40+lr.intercept,2))"
```

6 State the limit of that prediction: spend=40 is outside the observed range (10–33). Write down why extrapolating beyond your data is the analysis error the exam warns about.

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Lab 10 — Correlation, Regression and the Causation Trap — Procedure 3 of 3

THE TRAP — now look at staff versus revenue. The correlation is nearly as strong.

7

```
$ python3 -c "import pandas as pd;
   from scipy import stats;
   d=pd.read_csv('marketing.csv');
   r,p=stats.pearsonr(d.staff,d.revenue);
   print('staff vs revenue r =',round(r,4))"
```

8

Explain in writing: does hiring staff CAUSE revenue? Identify the confounding variable that drives both, and state the one sentence every analyst must be able to defend — correlation is not causation.

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Lab 10 — Verify Your Work

✅ Expected result

Across 36 months spend and revenue correlate at r ≈ 0.991 with R-squared ≈ 0.982, giving revenue ≈ 6.55 × spend + 54.1. Staff headcount ALSO correlates with revenue at r ≈ 0.970 — but growth over time drives both, so that second relationship is not causal.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

r is exactly 1.0

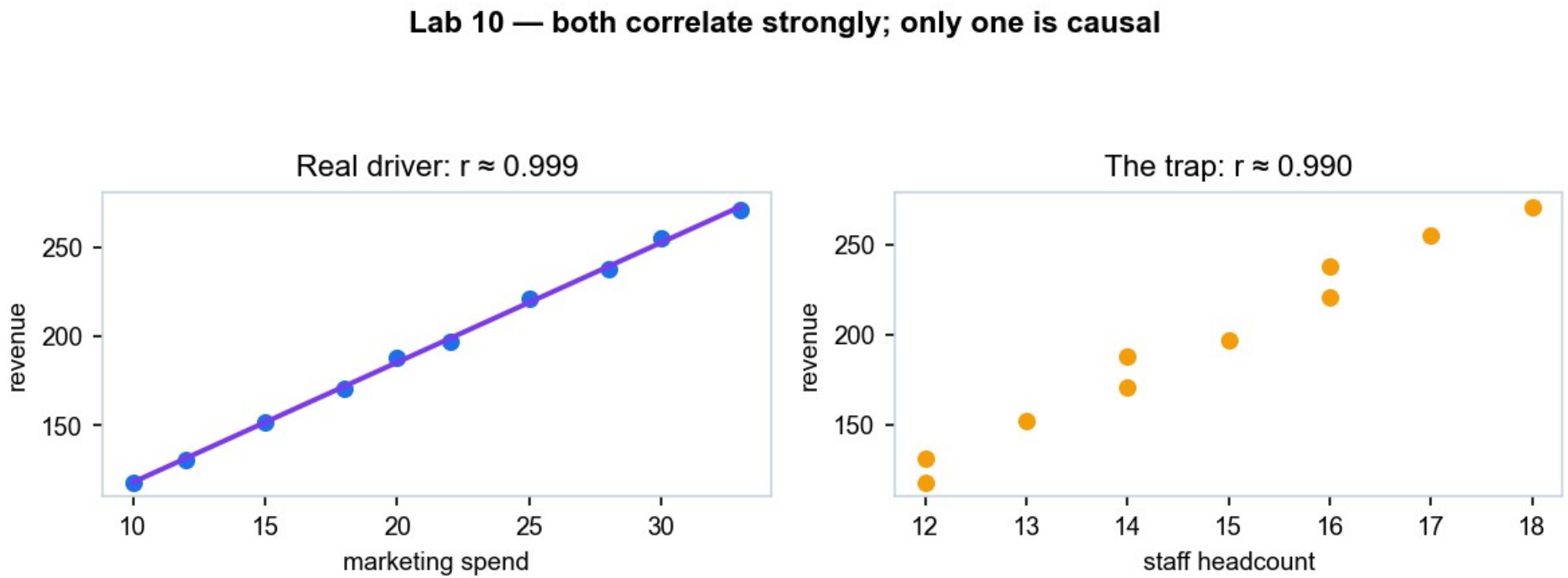
Perfect correlation means the data is synthetic and noise-free. Note that real business data never looks like this.

linregress has no attribute rvalue

You are on a very old scipy. Use r,p = stats.pearsonr(...) and square r yourself.

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Lab 10 — Expected Output



WHAT YOU SHOULD SEE

This is what your own run should produce. Across 36 months spend and revenue correlate at r ≈ 0.991 with R-squared ≈ 0.982, giving revenue ≈ 6.55 × spend + 54.1.

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Recap — Data Analysis

1. **Descriptive stats**
   Descriptive statistics summarise: mean, median and mode for centre; range, variance and SD for spread.
2. **Mean vs median**
   The mean is dragged by outliers and the median resists them — which one you report changes the story.
3. **Z-scores**
   Z-scores standardise distance from the mean, and $|z| > 3$ is the conventional outlier flag.
4. **Inferential stats**
   Inferential statistics generalise from a sample: t-tests, chi-square, correlation and regression.
5. **Hypothesis testing**
   State H0 and H1 before you test; $p < 0.05$ rejects the null; Type I is a false alarm, Type II a missed effect.
6. **Correlation**
   Correlation measures a pattern; causation requires a mechanism, time ordering and controlled confounders.
7. **Audience**
   Match the delivery to the audience — executives get the decision, technical experts get the method.

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Lunch Break

1 hour

04

DOMAIN 04 · 20% OF THE EXAM

Visualization and Reporting

Chart selection · maps · tables · design elements · dashboards · report types · validation

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Domain 4 — Key Concepts

1. **The question picks the chart**
   Chart choice follows the question: composition, comparison, distribution, relationship or trend over time.
2. **Time, category, distribution**
   Line charts carry time series, bar/column charts compare categories, histograms show distribution, scatter plots show relationship.
3. **Composition and hierarchy**
   Pie charts suit a few broad parts of a whole; treemaps suit hierarchical subcategories; heat maps encode magnitude in colour.
4. **Geographic data**
   Geographic data is shown as dot, filled (choropleth) or layered maps; ArcGIS, Power BI and Tableau all render them.
5. **Dashboards stay focused**
   A dashboard tracks a small set of decisions — filters, drillthrough and tooltips keep it simple rather than exhaustive.
6. **Report types**
   Reports are static, dynamic, real-time, operational, compliance, ad hoc or self-service — the type follows the audience and cadence.
7. **Validating accuracy**
   Reporting accuracy is validated by cross-validation, peer review, record-count checks, recalculation and data audits.

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USING THE APPROPRIATE VISUALIZATION

The question determines the chart. Not your preference.

Composition, comparison, distribution, relationship or trend — every business question falls into one of these five shapes, and each shape has a chart that fits it. Choose the chart last, after you know the question.

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CHART SELECTION

Which Chart Answers Which Question?

The question	The chart	Why
How is it trending over time?	Line chart	A continuous line reads as a sequence
Which category is biggest?	Bar / column chart	Length is the easiest visual comparison
What share of the whole?	Pie chart (few slices)	Angle shows parts of one whole
How is one variable spread?	Histogram	Bins reveal the shape of a distribution
Do two variables relate?	Scatter plot	Position on two axes exposes a pattern
Where is it happening?	Map (dot / filled / layered)	Geography is its own dimension
How does a total build up?	Waterfall chart	Shows each contribution to a change

WHEN IT MATTERS Get this wrong and the visual actively misleads — in Lab 11 you build a deliberately wrong chart to see it happen.

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VISUALIZATION TYPES

Specialist Chart Types

1

Tree map

Hierarchical data as nested rectangles sized by value. Label only the large categories.

2

Heat map

Colour encodes magnitude across two dimensions — excellent for spotting concentration.

3

Bubble chart

A scatter plot with a third variable encoded as the size of the point.

4

Combination chart

Two chart types on shared axes — e.g. revenue columns with a margin-percentage line.

5

Word cloud

Word frequency as text size. Eye-catching, but imprecise — never use it for a real measurement.

6

Infographic

A designed narrative combining data and graphics. Built for marketing, not for analysis.

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Geographic Visualisation

1

Dot map

Markers placed at specific coordinates — each dot is one event or location.

2

Filled (choropleth) map

Shading fills a bounded region by value. Beware: large regions look more important than they are.

3

Layered map

Two dimensions at once — e.g. shaded regions for revenue, sized dots for store count.

4

Tools

ArcGIS for serious geospatial work; Power BI and Tableau both render maps directly from postal or lat/long data.

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From Requirement to Delivered Report

1. **Define the audience**
   Who decides what?
2. **Agree the questions**
   What must it answer?
3. **Document the sources**
   Where does data come from?
4. **Build and validate**
   Draw it, then check it
5. **Deliver on cadence**
   Who gets it, how often?

THE DISTRIBUTION LIST

Agree who receives the report BEFORE you build it — the audience determines the level of detail and what may legally be shared.

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DOCUMENTATION ELEMENTS

Report and Dashboard Components

1

Report header

First page only — the title, the version number and the reporting period.

2

Page header / footer

Repeat on every page — field headings at the top, page numbers and references at the bottom.

3

Report footer

Last page only — summary information and credit to the authors.

4

Watermark

Signals handling restrictions: DRAFT, CONFIDENTIAL, INTERNAL USE ONLY.

5

Refresh date

When the DATA was last updated — the single most-forgotten and most-important element.

6

Print date

When this copy was produced. Different from the refresh date, and both matter in an audit.

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Report Types

1

Static / point-in-time

A fixed snapshot that does not update. The record of what was true at that moment.

2

Dynamic / real-time

Refreshes on access or on a schedule — always shows the current state.

3

Operational

The status of projects, products or processes, usually daily and detailed.

4

Compliance

Produced to satisfy a regulator. The format and cadence are usually prescribed for you.

5

Ad hoc

Generated once to answer a specific one-off question.

6

Self-service

The platform lets end users build their own — which shifts governance onto the data model.

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Dashboard Design Principles

1

One purpose

A dashboard tracks a small set of decisions. If it answers everything, it answers nothing.

2

Consistent colour

Never pick colours at random. One meaning per colour, applied identically on every panel.

3

Filters and sorting

Hardcoded, interactive, visual and date filters; natural order, multi-sort, top/bottom-N and custom sorts.

4

Drillthrough & tooltips

Keep the surface simple, and let the detail live one click away.

5

Labels and legends

Every axis labelled, every unit stated. An unlabelled number is not a finding.

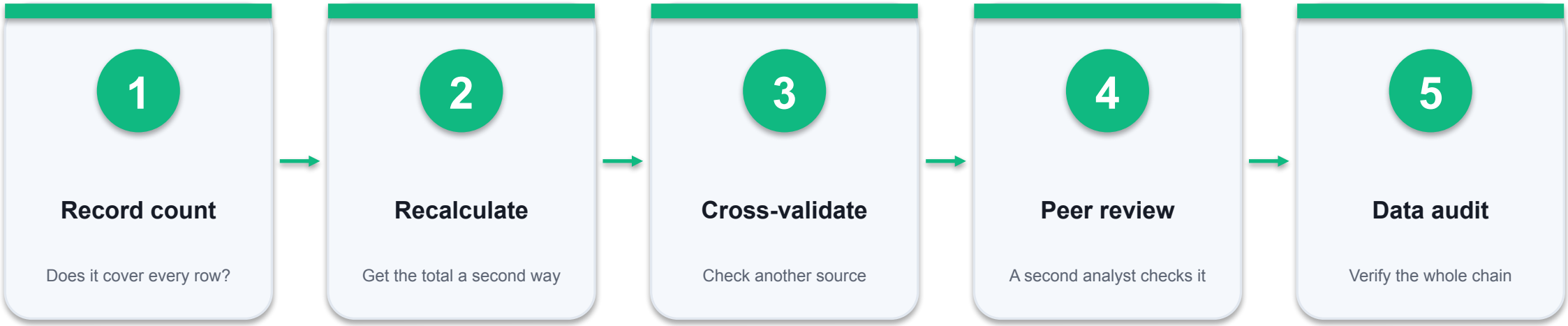
6

Test before deploy

Check the labels, test the filters, confirm permissions and licensing, then document it.

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Validating Reporting Accuracy



THE TEST OF A GOOD PROCESS

A validation process that has never caught an error has not been proven. In Lab 12 you plant an error deliberately to prove yours works.

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QUALITY CHECKS

Data Validation vs Data Verification

	Validation	Verification
Checks	The FORMAT and STRUCTURE of the data	The ACCURACY of the data
Question asked	Is it the right shape and type?	Is it actually true and correct?
Example	The date field contains a parseable date	The date matches the source system's record
When it runs	On entry and on import — often automated	On review — record counts, recalculation, peer review
Catches	Wrong types, invalid values, missing fields	Wrong joins, wrong formulas, transposed figures

WHEN IT MATTERS Both are required. Data can be perfectly valid in format and still be completely wrong in fact.

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Lab 11 — Choose the Right Chart: Five Questions, Five Chart Types

LAB 11

Create effective visuals: use charts, maps, tables and design elements (Domain 4); LO4 / K3 / A5.

Chart choice is not decoration — it is determined by the question being asked. You take ONE dataset and answer five different business questions from it, each demanding a different chart type, then build a deliberately wrong chart so you can articulate exactly why it misleads.

YOU'LL BUILD

Five correctly chosen charts (line, bar, pie, histogram, scatter) as PNG files, plus one annotated 'wrong chart' example.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas, matplotlib

DONE WHEN

Six PNG files exist, built from 24 rows covering 6 months × 4 regions. Each of the five correct charts answers its stated question, and your critique of the wrong chart explains that a pie destroys the sequence and makes a trend impossible to read.

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How Lab 11 Runs

1

Download this lab's dataset

step 1 of 9

2

Q2 'Which region sells most?' → BAR CHART, b

step 3 of 9

3

Q4 'How are order sizes distributed?' → HIST

step 5 of 9

4

Now build the WRONG chart on purpose

step 7 of 9

5

Write the critique of 06_wrong

step 9 of 9

YOU'LL PRODUCE

Five correctly chosen charts (line, bar, pie, histogram, scatter) as PNG files, plus one annotated 'wrong chart' example.

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Lab 11 — Choose the Right Chart: Five Questions, Five Chart Types — Procedure 1 of 3

1

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

```
$ mkdir -p ~/dataplus/lab11 && cd ~/dataplus/lab11;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-11-choose-the-right-chart-five-questions-five-chart-typ;
   for f in sales.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

Q1 'How is revenue trending?' → LINE CHART, because the x-axis is time.

```
$ python3 -c "import pandas as pd,matplotlib;matplotlib.use('Agg');import matplotlib.pyplot as
   plt;d=pd.read_csv('sales.csv');p=d.pivot_table(index='month',columns='region',values='revenue')
   .reindex(['Jan','Feb','Mar']);p.plot(marker='o');plt.title('Revenue Trend by Region');plt.ylabel('Revenue (SGD
   k)');plt.tight_layout();plt.savefig('01_line.png',dpi=120)"
```

3

Q2 'Which region sells most?' → BAR CHART, because you are comparing categories.

```
$ python3 -c "import pandas as pd,matplotlib;matplotlib.use('Agg');import matplotlib.pyplot as
   plt;d=pd.read_csv('sales.csv');d.groupby('region').revenue.sum().sort_values().plot(kind='barh',
   color='#1F6FEB');plt.title('Total Revenue by Region');plt.xlabel('Revenue (SGD
   k)');plt.tight_layout();plt.savefig('02_bar.png',dpi=120)"
```


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Lab 11 — Choose the Right Chart: Five Questions, Five Chart Types — Procedure 2 of 3

Q3 'What share does each region hold?' → PIE CHART, because it is parts of one whole (and only 3 slices).

4

```
$ python3 -c "import pandas as pd,matplotlib;matplotlib.use('Agg');import matplotlib.pyplot as
  plt;d=pd.read_csv('sales.csv');d.groupby('region').revenue.sum().plot(kind='pie',autopct='%1.1f%%',
  colors=['#1F6FEB','#10B981','#7C3AED']);plt.title('Revenue Share by
  Region');plt.ylabel('');plt.tight_layout();plt.savefig('03_pie.png',dpi=120)"
```

Q4 'How are order sizes distributed?' → HISTOGRAM, because you want the shape of one variable.

5

```
$ python3 -c "import pandas as pd,matplotlib;matplotlib.use('Agg');import matplotlib.pyplot as
  plt;d=pd.read_csv('sales.csv');d.orders.plot(kind='hist',bins=5,color='#F59E0B',edgecolor='white');
  plt.title('Distribution of Order Counts');plt.xlabel('Orders per
  region-month');plt.tight_layout();plt.savefig('04_hist.png',dpi=120)"
```

Q5 'Do more orders mean higher revenue?' → SCATTER PLOT, because you are testing a relationship.

6

```
$ python3 -c "import pandas as pd,matplotlib;matplotlib.use('Agg');import matplotlib.pyplot as
  plt;d=pd.read_csv('sales.csv');plt.scatter(d.orders,d.revenue,c='#7C3AED',s=80);plt.title('Orders vs
  Revenue');plt.xlabel('Orders');plt.ylabel('Revenue (SGD
  k)');plt.tight_layout();plt.savefig('05_scatter.png',dpi=120)"
```


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Lab 11 — Choose the Right Chart: Five Questions, Five Chart Types — Procedure 3 of 3

Now build the WRONG chart on purpose — a pie chart of a time series, which destroys the time ordering.

7

```
$ python3 -c "import pandas as pd,matplotlib;matplotlib.use('Agg');import matplotlib.pyplot as
  plt;d=pd.read_csv('sales.csv');d.groupby('month').revenue.sum().plot(kind='pie',autopct='%1.1f%%');
  plt.title('WRONG: revenue by month as a
  pie');plt.ylabel('');plt.tight_layout();plt.savefig('06_wrong.png',dpi=120)"
```

List the generated files and write one line per chart stating the question it answers.

8

```
$ ls -1 *.png
```

9

Write the critique of 06_wrong.png: what does a pie chart hide that a line chart shows?

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Lab 11 — Verify Your Work

✅ Expected result

Six PNG files exist, built from 24 rows covering 6 months × 4 regions. Each of the five correct charts answers its stated question, and your critique of the wrong chart explains that a pie destroys the sequence and makes a trend impossible to read.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

No display / tkinter error

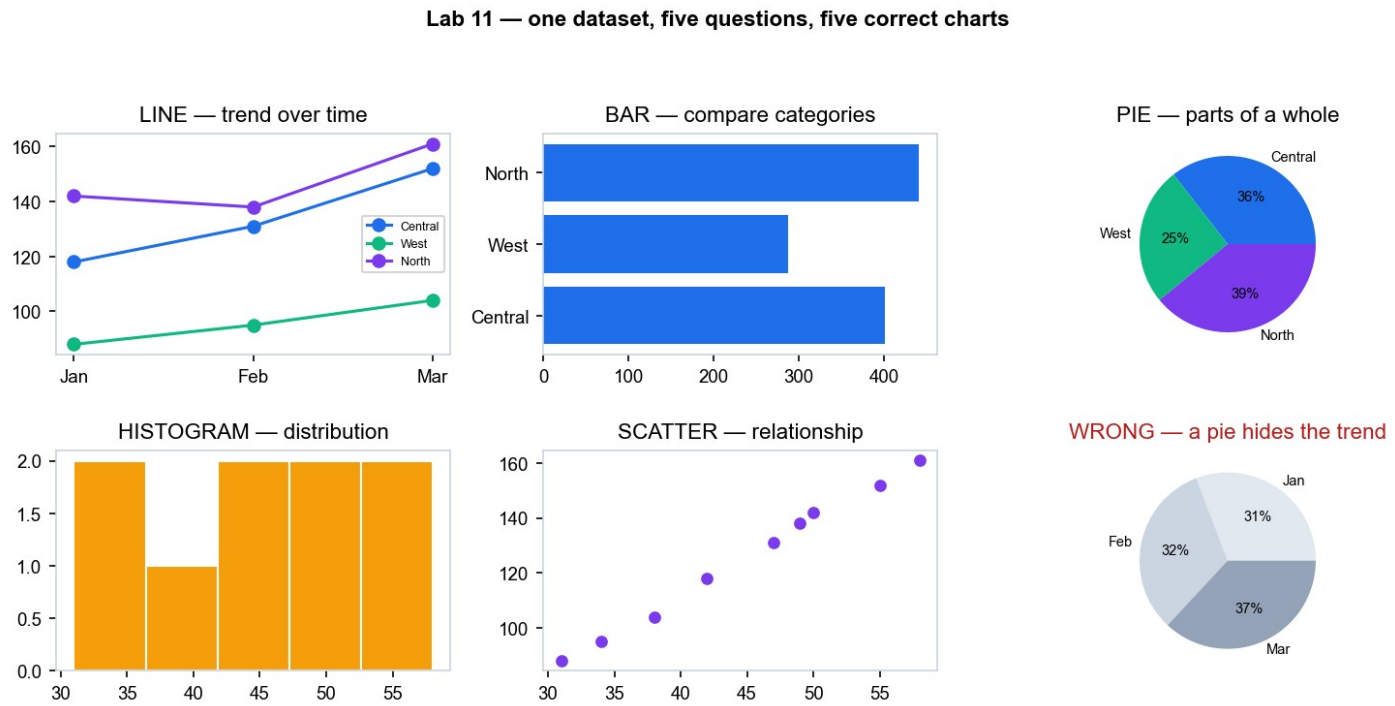
You must set matplotlib.use('Agg') BEFORE importing pyplot — headless terminals have no display.

The PNG is blank

You saved after calling plt.show() or a new figure. Call plt.savefig() before any clf/show, and use plt.close() between charts.

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Lab 11 — Expected Output



WHAT YOU SHOULD SEE

This is what your own run should produce. Six PNG files exist, built from 24 rows covering 6 months × 4 regions.

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Lab 12 — Build a KPI Dashboard and Validate Its Accuracy

LAB 12

Deliver reports: dashboards and summaries; validate reporting accuracy (Domain 4); LO4 / K3 / A5.

You assemble a four-panel executive dashboard from the integrated dataset, then run the validation checks the exam requires — record-count reconciliation, recalculation and cross-validation — and deliberately plant one error so you can prove your validation process actually catches it.

YOU'LL BUILD

A four-panel KPI dashboard PNG plus a signed validation checklist that catches a planted reporting error.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas, matplotlib

DONE WHEN

dashboard.png shows four panels built from 24 rows. Total revenue validates at 3021.3 by two independent methods. After the planted error the total jumps to 4456.8 and North reports 276.9% of target — an impossible figure your validation catches immediately. Restoring the value returns the total to 3021.3.

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How Lab 12 Runs

1

Download this lab's dataset

step 1 of 10

2

Build the four-panel dashboard in one script

step 3 of 10

3

VALIDATION 3

step 6 of 10

4

Re-run validation 2 and 3 and confirm your c

step 8 of 10

5

Sign off the validation checklist: record co

step 10 of 10

YOU'LL PRODUCE

A four-panel KPI dashboard PNG plus a signed validation checklist that catches a planted reporting error.

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Lab 12 — Build a KPI Dashboard and Validate Its Accuracy — Procedure 1 of 4

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

1

```
$ mkdir -p ~/dataplus/lab12 && cd ~/dataplus/lab12;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-12-build-a-kpi-dashboard-and-validate-its-accuracy;
   for f in kpi.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

Compute the headline KPIs first — you must know the true numbers BEFORE you draw anything.

2

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('kpi.csv');
   print('total revenue',d.revenue.sum());
   print('total orders',d.orders.sum());
   print('avg order value',round(d.revenue.sum()*1000/d.orders.sum(),2));
   print('vs target',round(d.revenue.sum()*100/d.target.sum(),1),'%')"
```

Build the four-panel dashboard in one script.

3

```
$ cat > dashboard.py <<'EOF'   # full script in the Learner Guide / lab README
```


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Lab 12 — Build a KPI Dashboard and Validate Its Accuracy — Procedure 2 of 4

4

VALIDATION 1 — record count. Does the dashboard cover every source row?

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('kpi.csv');
   print('source rows',len(d));
   print('rows charted',d.groupby(['month','region']).ngroups)"
```

5

VALIDATION 2 — recalculate the headline total a second, independent way.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('kpi.csv');
   a=d.revenue.sum();
   b=sum(d.groupby('region').revenue.sum());
   print('method A',a,'method B',b,'MATCH' if a==b else 'MISMATCH')"
```

6

VALIDATION 3 — cross-validate the % of target figure against a hand calculation.

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('kpi.csv');
   print('Central', round(d[d.region=='Central'].revenue.sum()/d[d.region=='Central'].target.sum()*100,1),'% expected 102.9')"
```


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Lab 12 — Build a KPI Dashboard and Validate Its Accuracy — Procedure 3 of 4

Now PLANT AN ERROR — corrupt one revenue value and rebuild the dashboard.

7

```
$ sed -i 's/Mar,North,159.5/Mar,North,1595/' kpi.csv && python3 dashboard.py
```

Re-run validation 2 and 3 and confirm your checks CATCH the planted error.

8

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('kpi.csv');
   print('total now',round(d.revenue.sum(),1),'(was 3021.3)');
   print('North % of target',round(d[d.region=='North'].revenue.sum()/d[d.region=='North'].target.sum()*100,1),'%')"
```

Restore the correct value and confirm the dashboard returns to its validated state.

9

```
$ sed -i 's/Mar,North,1595/Mar,North,159.5/' kpi.csv && python3 dashboard.py && python3 -c "import pandas as pd;
   print('total',pd.read_csv('kpi.csv').revenue.sum())"
```


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Lab 12 — Build a KPI Dashboard and Validate Its Accuracy — Procedure 4 of 4

10 Sign off the validation checklist: record count, recalculation, cross-validation, and visual review.

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Lab 12 — Verify Your Work

✅ Expected result

dashboard.png shows four panels built from 24 rows. Total revenue validates at 3021.3 by two independent methods. After the planted error the total jumps to 4456.8 and North reports 276.9% of target — an impossible figure your validation catches immediately. Restoring the value returns the total to 3021.3.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

Panels overlap or the title is cut

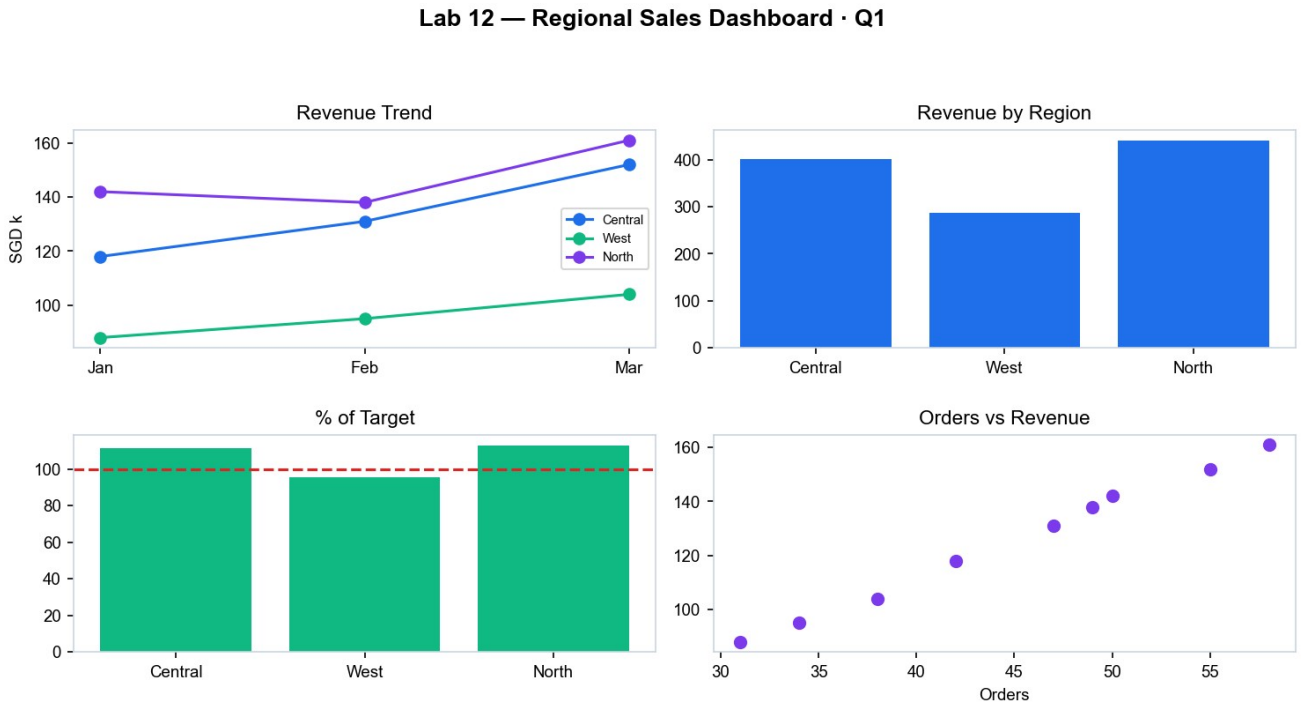
Use plt.tight_layout(rect=[0,0,1,0.95]) so the suptitle gets its own space.

sed did not change anything

sed needs a BYTE-EXACT match. Run grep 'Mar,North' kpi.csv and copy the value from the output — 159.5 is not the same string as 159.50 or 161.

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Lab 12 — Expected Output



WHAT YOU SHOULD SEE

This is what your own run should produce. dashboard.png shows four panels built from 24 rows. Total revenue validates at 3021.3 by two independent methods.

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Recap — Visualization and Reporting

1. **Question first**
   The business question determines the chart: composition, comparison, distribution, relationship or trend.
2. **Chart selection**
   Line for time, bar for categories, pie for a few parts of a whole, histogram for distribution, scatter for relationship.
3. **Maps & magnitude**
   Maps come as dot, filled (choropleth) or layered; heat maps and tree maps encode magnitude and hierarchy.
4. **Audience & sharing**
   Agree the audience and the distribution list before you build — they set the detail level and what may be shared.
5. **Report types**
   Reports are static, dynamic, operational, compliance, ad hoc or self-service, and each carries standard elements.
6. **Dashboard design**
   A good dashboard has one purpose, consistent colour, labelled axes and detail hidden behind drillthrough.
7. **Validation**
   Validate by record count, recalculation, cross-validation, peer review and audit — and prove the process catches errors.

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Tea Break

15 minutes

05

DOMAIN 05 · 14% OF THE EXAM

Data Governance, Quality and Controls

Documentation · versioning · lineage · compliance · retention · privacy · encryption · masking · quality assurance

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WHAT THIS DOMAIN COVERS

Domain 5 — Key Concepts

1. **Governance spans the lifecycle**
   Data governance keeps data high-quality and controlled across its full lifecycle: creation, storage, use, archive and destruction.
2. **Roles split accountability**
   Data roles separate accountability: data owner, data steward, data custodian and privacy officer.
3. **Classification and data types**
   Data is classified by sensitivity (public, internal, sensitive, confidential/restricted) and tagged by type (PII, PHI, PIFI, IP).
4. **Compliance obligations**
   Regulations impose retention, audit and sovereignty duties — Singapore's PDPA, the EU GDPR, HIPAA and SOX are the common examples.
5. **Layered protection**
   Protection strategies layer access control (role-based, group-based), encryption at rest/in transit/in use, masking and de-identification.
6. **Lineage and version control**
   Data lineage, a data dictionary and version control make a pipeline auditable and reproducible.
7. **Continuous quality assurance**
   Quality assurance runs continuously: profiling, automated validation on entry, monitoring, and testing against agreed quality dimensions.

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DATA GOVERNANCE

Governance is a set of decisions, not a document.

Data governance is an organisation's capability to ensure high-quality data exists across the complete data lifecycle, with controls implemented to support its business objectives. Every control is somebody's decision, written down and enforced.

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The Data Lifecycle



WHY IT DRIVES POLICY

Every governance policy you write must say what happens at each of these five stages — otherwise there is a stage where nobody is accountable.

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Data Roles and Accountability

1

Data owner

A senior executive accountable for the confidentiality, integrity and availability of the data asset.

2

Data steward

Ensures data is properly labelled, identified, collected and stored. The day-to-day quality role.

3

Data custodian

Manages the SYSTEM the data lives on — the technical operator of storage and backup.

4

Privacy officer

Oversees privacy-related data, and owns minimisation, sovereignty, retention and destruction policy.

5

Why separate them

Separation of duties: the person who authorises access should not be the person who grants it.

6

In practice

In Lab 14 you build the access matrix that assigns each of these roles a specific level of access.

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Data Classification — Commercial and Government

Level	Commercial sector	Government sector
Lowest	Public — the front-facing website	Unclassified — public information
Low-mid	Sensitive — financial data	Controlled Unclassified (CUI) — medical records
Mid	Private — personnel records	Confidential — trade secrets
High	Confidential — intellectual property	Secret — military plans
Highest	Restricted — regulated or contractual	Top Secret — weapon blueprints

WHEN IT MATTERS Classification sets the sensitivity. A separate DATA TYPE tag — PII, PHI, PIFI, IP — says what kind of data it is.

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COMPLIANCE REQUIREMENTS

Compliance and Regulation

1

Regulation vs compliance

A regulation is a rule backed by law. Compliance is demonstrating that you meet it.

2

Singapore PDPA

Governs collection, use and disclosure of personal data, and mandates breach notification.

3

GDPR (EU)

Rights of access, rectification, erasure and portability — applies to EU residents' data wherever it is held.

4

HIPAA / SOX

US health information privacy; and financial reporting controls with retention obligations.

5

Data sovereignty

The jurisdiction whose law governs the data — usually determined by where it is physically stored.

6

Audits

Periodic independent verification that the controls you documented are actually operating.

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Retention, Preservation and Destruction

1

Data retention

Keeping data for a defined period to satisfy business policy and applicable law. Consult legal counsel.

2

Data preservation

Holding data for a specific purpose OUTSIDE the retention policy — for example, a litigation hold.

3

Data removal

Deleting data or making it inaccessible. Appropriate only for the least sensitive data.

4

Data destruction

Deleting the data AND destroying the underlying medium so it cannot be reconstructed.

5

Data sanitisation

The verification step that PROVES the data was wiped and is no longer recoverable.

6

Document it

An undocumented retention period is itself an audit finding — record the period and its rationale.

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Protection Strategies Compared

Technique	What it does	Reversible?	Analysis still possible?
Access control	Restricts WHO can see the data	n/a	Yes, for authorised users
Encryption	Scrambles data into ciphertext	Yes, with the key	Only after decryption
Masking	Hides part of the value (S****A)	No	Partially — recognition only
De-identification	Removes the identifying fields	No	Yes, on remaining fields
Pseudonymisation	Replaces the ID with a surrogate key	Only with the mapping	Yes — joins still work

WHEN IT MATTERS You apply masking, de-identification and pseudonymisation to the same dataset in Lab 13 — and prove the analysis survives.

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Encryption and Access Control

1

Data at rest

Stored as ciphertext on disk; re-encrypted on every write.

2

Data in transit

Encrypted while moving between systems — TLS/HTTPS is the baseline expectation.

3

Data in use

Being processed in memory. The hardest state to protect, and the target of modern attacks.

4

Read/write permissions

The most basic control: who may read, and who may also change.

5

Role-based access (RBAC)

Access assigned by JOB FUNCTION, not by individual. Scales, and survives staff changes.

6

User-group permissions

Access granted by group membership. Simple, but drifts unless it is reviewed regularly.

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Documentation, Versioning and Lineage

1

Data lineage

The traceable path from origin through every transformation to the final report. Answers 'where did this number come from?'

2

Data dictionary

Defines every field: its meaning, type, valid values, owner and source. The reference everyone shares.

3

Version control

Tracks changes to queries, models and reports over time, so any figure can be reproduced later.

4

Entity relationship models

Conceptual (what exists), logical (fields and relationships), physical (actual tables and types).

5

Data constraints

Integrity rules limiting what may enter a column — the schema enforcing quality automatically.

6

Record linkage

Matching records across datasets — with link restrictions preventing protected sets being combined.

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The Data Quality Dimensions

1

Completeness

No required data is missing. Measured as the null rate on mandatory fields.

2

Accuracy

The values are actually correct — they match the real-world fact they describe.

3

Consistency

The same thing is recorded the same way everywhere. 'SG', 'Singapore' and 'sg' break this.

4

Uniqueness

No unintended duplicates. The primary key must identify exactly one record.

5

Validity

Values conform to the defined format, type, range and domain.

6

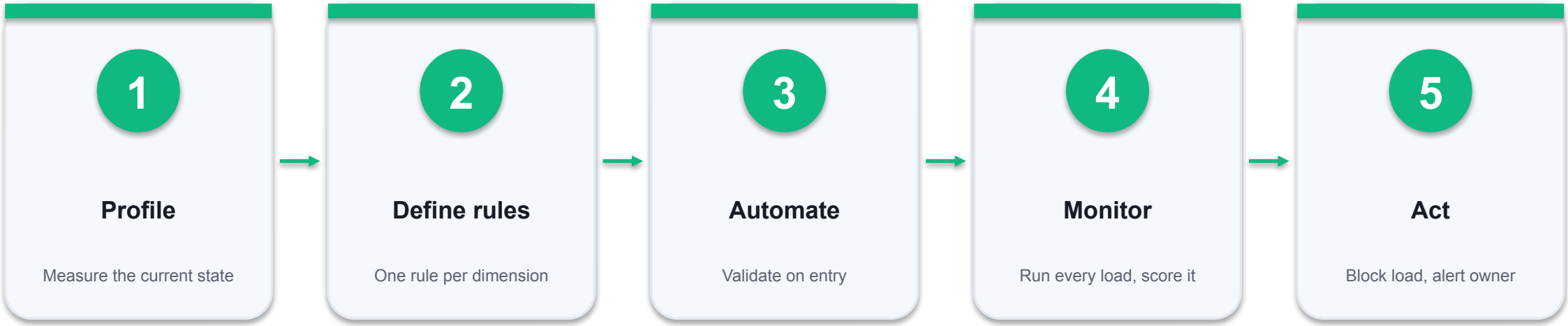
Timeliness

The data is current enough for the decision it supports. A stale figure is a wrong figure.

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PROFILE · MONITOR · TEST

Quality Assurance in Operation



WHAT MAKES IT MONITORING

A report tells you the data was bad. A non-zero exit code STOPS the bad data loading — that is the difference you build in Lab 15.

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Master Data Management

1

Master data

The golden record — the single authoritative version of a core entity such as a customer or product.

2

MDM

The discipline of creating and maintaining that golden record across every system that holds a copy.

3

How golden records are built

Consolidate duplicate fields, fill gaps from other systems, and standardise the field formats.

4

Why it matters

Without it, Finance, Sales and Support each report a different customer count — all defensible, all different.

5

MDM and governance

They work together: governance sets the rules, MDM enforces them on the core entities.

6

Tooling

Informatica and Reltio are common enterprise MDM platforms.

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Lab 13 — Classify, Mask and De-Identify a Dataset (PDPA/GDPR)

LAB 13

Compare privacy and protection strategies: access control, encryption and masking (Domain 5); LO5 / A4.

You receive a customer extract containing NRIC numbers, emails and salaries. You classify every column by sensitivity, then apply the three protection techniques the exam distinguishes — masking, de-identification and pseudonymisation via a surrogate index field — and prove the analytical value survives the treatment.

YOU'LL BUILD

A classification matrix for every column plus three protected versions of the dataset (masked, de-identified, pseudonymised) with the analysis still working.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas, hashlib

DONE WHEN

Across 40 records masking shows the NRIC as S****G form. De-identification drops name, nric and email. Pseudonymisation gives a stable 12-character key, and the departmental salary averages (Operations 4006.25, Sales 4518.75, Technology 5431.25, Finance 4612.50, Marketing 4268.75) are identical to those computed on the raw data.

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How Lab 13 Runs

1. **Download this lab's dataset**
   step 1 of 8
2. **Apply MASKING**
   step 3 of 8
3. **Apply PSEUDONYMISATION**
   step 5 of 8
4. **Test the re-identification risk: group by po**
   step 7 of 8
5. **Save the release-ready pseudonymised extract**
   step 8 of 8

YOU'LL PRODUCE

A classification matrix for every column plus three protected versions of the dataset (masked, de-identified, pseudonymised) with the analysis still working.

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Lab 13 — Classify, Mask and De-Identify a Dataset (PDPA/GDPR) — Procedure 1 of 3

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

1

```
$ mkdir -p ~/dataplus/lab13 && cd ~/dataplus/lab13;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-13-classify-mask-and-de-identify-a-dataset-pdpa-gdpr;
   for f in customers.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

2

CLASSIFY first. For each column write down its classification (public / internal / sensitive / confidential) and its data type tag (PII, PIFI, none). NRIC is confidential PII; salary is sensitive PIFI; dept is internal.

Apply MASKING — the value stays partly visible so it can still be recognised by an authorised human.

3

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('customers.csv');
   d['nric']=d.nric.str[0]+'****'+d.nric.str[-1];
   d['email']=d.email.str.replace(r'(.).*(@.*)',r'\1****\2',regex=True);
   print(d[['cust_id','name','nric','email']])"
```


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Lab 13 — Classify, Mask and De-Identify a Dataset (PDPA/GDPR) — Procedure 2 of 3

Apply DE-IDENTIFICATION — the identifying columns are removed outright.

4

```
$ python3 -c "import pandas as pd;
   d=pd.read_csv('customers.csv');
   deid=d.drop(columns=['name','nric','email']);
   print(deid)"
```

Apply PSEUDONYMISATION — replace the identifier with a non-reversible surrogate INDEX FIELD.

5

```
$ python3 -c "import pandas as pd,hashlib;
   d=pd.read_csv('customers.csv');
   d['subject_key']=d.nric.map(lambda v: hashlib.sha256(v.encode()).hexdigest()[:12]);
   print(d[['cust_id','subject_key','dept','salary']])"
```

Prove the ANALYSIS STILL WORKS on the protected data — this is the whole point of the technique.

6

```
$ python3 -c "import pandas as pd,hashlib;
   d=pd.read_csv('customers.csv');
   d['subject_key']=d.nric.map(lambda v: hashlib.sha256(v.encode()).hexdigest()[:12]);
   print(d.groupby('dept').salary.mean().round(2))"
```


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Lab 13 — Classify, Mask and De-Identify a Dataset (PDPA/GDPR) — Procedure 3 of 3

7 **Test the re-identification risk: group by postal code and find the SMALLEST group. Any group of 1 is uniquely identifying. Discuss why small groups defeat de-identification (the k-anonymity…**

```
$ python3 -c "import pandas as pd;d=pd.read_csv('customers.csv');print(d.groupby('postal').size())"
```

8 **Save the release-ready pseudonymised extract and note who may access it under which role.**

```
$ python3 -c "import pandas as pd,hashlib;d=pd.read_csv('customers.csv');d['subject_key']=d.nric.map(lambda v:
   hashlib.sha256(v.encode()).hexdigest()[:12]);d[['subject_key','postal','dept',
   'salary']].to_csv('customers_release.csv',index=False)" && cat customers_release.csv
```


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Lab 13 — Verify Your Work

✅ Expected result

Across 40 records masking shows the NRIC as S****G form. De-identification drops name, nric and email. Pseudonymisation gives a stable 12-character key, and the departmental salary averages (Operations 4006.25, Sales 4518.75, Technology 5431.25, Finance 4612.50, Marketing 4268.75) are identical to those computed on the raw data.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

The email regex did not mask

Confirm regex=True is passed to str.replace and the backreferences use \1 and \2.

The hash changes between runs

SHA-256 is deterministic — if it changes, your input has stray whitespace. Strip it first.

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Lab 14 — Assess Data-Leakage Risk and Set Access Controls

LAB 14

Summarize compliance requirements; compare privacy and protection strategies (Domain 5); LO5 / A4.

Governance is a set of decisions, not a document. You use the Cybersecurity Threat Simulator's data-leakage risk estimator to score a realistic set of handling practices, then translate the score into a concrete role-based access matrix and a retention schedule for the Lab 13 dataset.

YOU'LL BUILD

A completed risk assessment with a before/after score, a role-based access control matrix, and a retention & destruction schedule.

TOOLCHAIN

Cybersecurity Threat Simulator (https://alfredang.github.io/cybersecuritysimulator/), the Lab 13 dataset

DONE WHEN

Your risk score drops from Critical to Medium or lower once encryption, RBAC, review and retention are enabled. Your access matrix gives the Analyst the pseudonymised release ONLY, never the raw NRIC.

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How Lab 14 Runs

1

Open the Cybersecurity Threat Simulator and

step 1 of 8

2

Now switch on the controls one at a time

step 3 of 8

3

Open the Password Strength Analyzer and test

step 5 of 8

4

Write the RETENTION SCHEDULE: how long is ea

step 7 of 8

5

Map each decision to its compliance driver

step 8 of 8

YOU'LL PRODUCE

A completed risk assessment with a before/after score, a role-based access control matrix, and a retention & destruction schedule.

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DOMAIN 05 · LAB 14

Lab 14 — Assess Data-Leakage Risk and Set Access Controls — Procedure 1 of 2

1. Open the Cybersecurity Threat Simulator and go to the Data Leakage Risk Estimator.
2. Set the toggles to describe a BAD baseline: no encryption, shared logins, no access review, data kept forever. Record the risk score and its level (Critical / High / Medium).
3. Now switch on the controls one at a time — encryption at rest, role-based access, periodic review, defined retention. Record how much each single control moves the score.
4. Note which single control reduced the risk most. That is where governance effort pays off first.

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Lab 14 — Assess Data-Leakage Risk and Set Access Controls — Procedure 2 of 2

5. Open the Password Strength Analyzer and test a weak versus a strong credential — this is the access control layer protecting everything you built in Lab 13.

6. Build the ACCESS MATRIX for the Lab 13 dataset. For each of the four roles (Data Owner, Data Steward, Data Custodian, Analyst) decide access to: raw NRIC, masked extract, pseudonymised release, salary.

7. Write the RETENTION SCHEDULE: how long is each artefact kept, what triggers destruction, and which method is used (removal vs destruction vs sanitisation)?

8. Map each decision to its compliance driver — Singapore PDPA for the personal data, and the organisation's own audit requirement for the retention log.

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Lab 14 — Verify Your Work

✅ Expected result

Your risk score drops from Critical to Medium or lower once encryption, RBAC, review and retention are enabled. Your access matrix gives the Analyst the pseudonymised release ONLY, never the raw NRIC.

IF IT DOESN'T WORK

The score does not change

Some toggles only affect certain threat categories. Change one at a time and re-read the score after each.

Unsure which role gets what

Apply least privilege: the Analyst needs the analytical columns, never the identifiers. Only the Data Owner authorises raw access.

Retention period unclear

Where no statutory period applies, set it from business need and document the rationale — an undocumented period is itself an audit finding.

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Lab 15 — Automated Quality Assurance: Profile, Rule, Monitor

LAB 15

Implement quality assurance: profiling, monitoring and testing for data quality; explain data management practices (Domain 5); LO5 / A4.

The capstone governance lab. You write an automated data-quality suite that tests a daily feed against explicit rules across the exam's quality dimensions — completeness, accuracy, consistency, uniqueness and validity — then run it against a clean file and a broken one so it proves it can actually fail.

YOU'LL BUILD

A reusable dq_check.py suite that scores five quality dimensions, exits non-zero on failure, and produces a dated quality report.

TOOLCHAIN

Killercoda Ubuntu, Python 3, pandas

DONE WHEN

feed_good.csv (40 rows) scores 5/5 and exits 0. feed_bad.csv (40 rows) fails ALL FIVE dimensions — 2 nulls, 2 duplicate order_ids, 2 negative amounts, 2 invalid status values and 1 unparseable date — so it scores 0/5, exits 1, and the 'LOAD BLOCKED' branch fires.

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How Lab 15 Runs

1

Download this lab's dataset

step 1 of 7

2

Write the quality suite

step 2 of 7

3

Run it against the BROKEN feed

step 4 of 7

4

Save a dated quality report so you build a q

step 6 of 7

5

Add the lineage note: record the source, the

step 7 of 7

YOU'LL PRODUCE

A reusable dq_check.py suite that scores five quality dimensions, exits non-zero on failure, and produces a dated quality report.

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DOMAIN 05 · LAB 15

Lab 15 — Automated Quality Assurance: Profile, Rule, Monitor — Procedure 1 of 3

Download this lab's dataset. The same files ship in the course repo under this lab's data/ folder.

1

```
$ mkdir -p ~/dataplus/lab15 && cd ~/dataplus/lab15;
   R=https://raw.githubusercontent.com/tertiarycourses;
   B=$R/TGS-2024049212-CompTIA-Certified-Data-Training/main/labs;
   D=lab-15-automated-quality-assurance-profile-rule-monitor;
   for f in feed_good.csv feed_bad.csv; do curl -fsSO $B/$D/data/$f || echo FAILED $f; done; ls -l
```

Write the quality suite — one function per quality dimension.

2

```
$ cat > dq_check.py <<'EOF'   # full script in the Learner Guide / lab README
```

Run the suite against the GOOD feed — it must pass all five and exit 0.

3

```
$ python3 dq_check.py feed_good.csv; echo "exit code: $?"
```


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Lab 15 — Automated Quality Assurance: Profile, Rule, Monitor — Procedure 2 of 3

4

Run it against the BROKEN feed — it must fail four dimensions and exit 1.

```
$ python3 dq_check.py feed_bad.csv; echo "exit code: $?"
```

5

This exit code is what makes it MONITORING rather than a report — a scheduler can now block a bad load.

```
$ python3 dq_check.py feed_bad.csv > /dev/null 2>&1 && echo 'LOAD APPROVED' || echo 'LOAD BLOCKED - do not
   ingest'
```

6

Save a dated quality report so you build a quality history, not just a snapshot.

```
$ python3 dq_check.py feed_good.csv > dq_report_$(date +%Y%m%d).txt; ls -1 dq_report_*.txt
```


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Lab 15 — Automated Quality Assurance: Profile, Rule, Monitor — Procedure 3 of 3

7

Add the lineage note: record the source, the owner, the rule version and the run date. This is the documentation half of the exam's data-management objective.

```
$ printf 'source: orders feed (daily)\nowner: Sales Ops\nsteward: Data Quality team\nrules version: 1.0\nrun:
   %s\n' "$(date +%F)" > lineage.txt && cat lineage.txt
```


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Lab 15 — Verify Your Work

✅ Expected result

feed_good.csv (40 rows) scores 5/5 and exits 0. feed_bad.csv (40 rows) fails ALL FIVE dimensions — 2 nulls, 2 duplicate order_ids, 2 negative amounts, 2 invalid status values and 1 unparseable date — so it scores 0/5, exits 1, and the 'LOAD BLOCKED' branch fires.

IF IT DOESN'T WORK

The CSV contains '404: Not Found'

curl wrote the error page into the file. Confirm the BASE URL, re-run with -fsSO so curl fails loudly, or copy the files from the repo folder you cloned.

The heredoc breaks on the f-strings

Ensure you used <<'EOF' with the quotes — that stops the shell expanding anything inside.

Exit code is always 0

You ran it inside another command. Test with python3 dq_check.py feed_bad.csv; echo $? on its own line.

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Recap — Data Governance, Quality and Controls

1. **Lifecycle**
   Governance ensures quality data across the full lifecycle: creation, storage, use, archive and destruction.
2. **Roles**
   Owner, steward, custodian and privacy officer separate accountability — no single person holds every duty.
3. **Classification**
   Classification sets sensitivity; a data-type tag (PII, PHI, PIFI, IP) says what kind of data it is.
4. **Compliance**
   PDPA, GDPR, HIPAA and SOX impose retention, audit and sovereignty duties you must be able to evidence.
5. **Layered protection**
   Access control, encryption, masking, de-identification and pseudonymisation are layered, not alternatives.
6. **Lineage**
   Lineage, a data dictionary and version control are what make a number reproducible and auditable.
7. **Quality assurance**
   Quality assurance runs continuously: profile, define rules, automate validation, monitor and act on failure.

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WRAP-UP

Tools, Revision and Next Steps

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Data Analytic Tools — Languages

1

SQL

The query language for relational databases. Non-negotiable for any data role.

2

Python

General-purpose, with pandas, NumPy, matplotlib and scikit-learn — the dominant analysis stack.

3

R

Purpose-built for statistical computing and graphics; strong among statisticians and researchers.

4

Excel / Power Query

Still the most widely used analysis tool on earth. Pivot tables, formulas and Power Query transforms.

5

Notebooks

Jupyter and Colab combine code, output and narrative in one reproducible document.

6

Version control

Git tracks every change to your queries and scripts, so any result can be reproduced.

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BI, Visualisation and Statistical Platforms

1

Power BI

Microsoft's interactive BI product — Power Query for prep, DAX for measures, dashboards for delivery.

2

Tableau

Leading visual-analysis tool, strong at exploratory visualisation and mapping.

3

Qlik / Domo / MicroStrategy

Enterprise BI platforms combining data sources, dashboards and distribution.

4

AWS QuickSight / Oracle Analytics

Cloud-native BI delivering dashboards over cloud data warehouses.

5

SAS · SPSS · Stata · Minitab

Established statistical packages used for advanced and regulated analysis.

6

SSRS · Crystal Reports

Paginated, pixel-perfect operational and compliance reporting.

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LEARNING OUTCOMES

What You Achieved

1

LO1 · Integrate datasets

Built a normalised schema with enforced referential integrity and integrated three sources with SQL joins (Labs 1, 7).

2

LO2 · Mine for trends

Profiled a dirty dataset, parsed messy fields with validated regex, and derived new structured fields (Labs 4, 5, 6).

3

LO3 · Analyse statistically

Computed descriptive statistics, ran a t-test with p-values, and fitted a regression model (Labs 8, 9, 10).

4

LO4 · Communicate visually

Matched five chart types to five questions and built a validated KPI dashboard (Labs 11, 12).

5

LO5 · Link the variables

Classified and protected sensitive data, assessed leakage risk, and automated quality monitoring (Labs 13, 14, 15).

6

Certification ready

Covered all five CompTIA Data+ (DA0-001) exam domains at their examined weightings.

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exams.tertiaryinfotech.com

Practice Exam — Before You Certify

Attempt the official practice exam before you sit the real CompTIA Data+ certification.

CompTIA Data+ (DA0-001) practice exams: **https://exams.tertiaryinfotech.com/practice-exams/comptia/comptia-data-plus**

6 FULL EXAMS

The bundle covers all five domains at their examined weightings.

TIMED PRACTICE

Sit them under exam conditions to build your pacing before the real thing.

REVIEW EVERY MISS

Read the explanation for every wrong answer — that is where the learning is.

THEN BOOK

Book the real DA0-001 exam once you are consistently scoring above your target.

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NEXT STEPS

Continue Your Learning

1

Sit the certification

Book the CompTIA Data+ (DA0-001) exam while the material is fresh.

2

Redo every lab

Repeat each lab from a blank terminal until the workflow is automatic.

3

Apply it at work

Profile one real dataset from your own job using the Lab 4 procedure this week.

4

Build a portfolio

Publish a cleaned dataset, an analysis and a dashboard you can show an employer.

5

Go deeper on SQL

Joins, window functions and query plans repay every hour you invest.

6

Learn a BI platform

Pick Power BI or Tableau and rebuild your Lab 12 dashboard in it.

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CONTINUE WITH TERTIARY INFOTECH

Recommended Courses

1. WSQ - CompTIA Certified Security+ Training
2. WSQ - CompTIA Certified Network+ Training
3. WSQ - CompTIA Certified Linux+ Training
4. WSQ - CompTIA Certified Server+ Training
5. WSQ - CompTIA Certified A+ Training (Core 1 and Core 2)

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WE'RE HERE TO HELP

Support

- If you have any enquiries during or after the class, you can contact us below.
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- Website: www.tertiarycourses.com.sg
- LMS / TMS: https://lms-tms.tertiaryinfotech.com

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FINAL ASSESSMENT

Assessment

1

Written Assessment (SAQ)

1 hour · open book · short-answer knowledge questions across all five domains.

2

Practical Performance (PP)

1 hour · open book · scenario-based hands-on data tasks mapped to LO1–LO5.

3

Digital attendance

Take the Assessment digital attendance (TRAQOM) before you start.

4

Submit on the LMS

Upload your completed answers at https://lms-tms.tertiaryinfotech.com/

5

Open book means

Course slides, the Learner Guide and other approved materials only.

6

Result

You are assessed Competent (C) or Not Yet Competent (NYC) on each instrument.

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ON ASSESSMENT DAY

Assessment Flow

1

TRAQOM attendance

Scan the QR on the LMS

2

Assessment attendance

Scan the SSG QR

3

Sit WA then PP

Open book · 1 hour each

4

Submit on the LMS

Upload your answers

5

Sign the record

Assessment Summary Record

REMEMBER

All five steps are mandatory for WSQ funding — missing the TRAQOM digital attendance or the course feedback survey can invalidate your claim.

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code generated from TPG.

3

Scan with Singpass

Scan the QR code from your Singpass App and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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HAPPY ANALYSING

Thank You!

You can now integrate, prepare, analyse, visualise and govern data — and you are ready to sit the CompTIA Data+ (DA0-001) certification exam.

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